4.3 Exercises

5 Corrective Maintenance (CM) Work

5.1 CM Work Overview

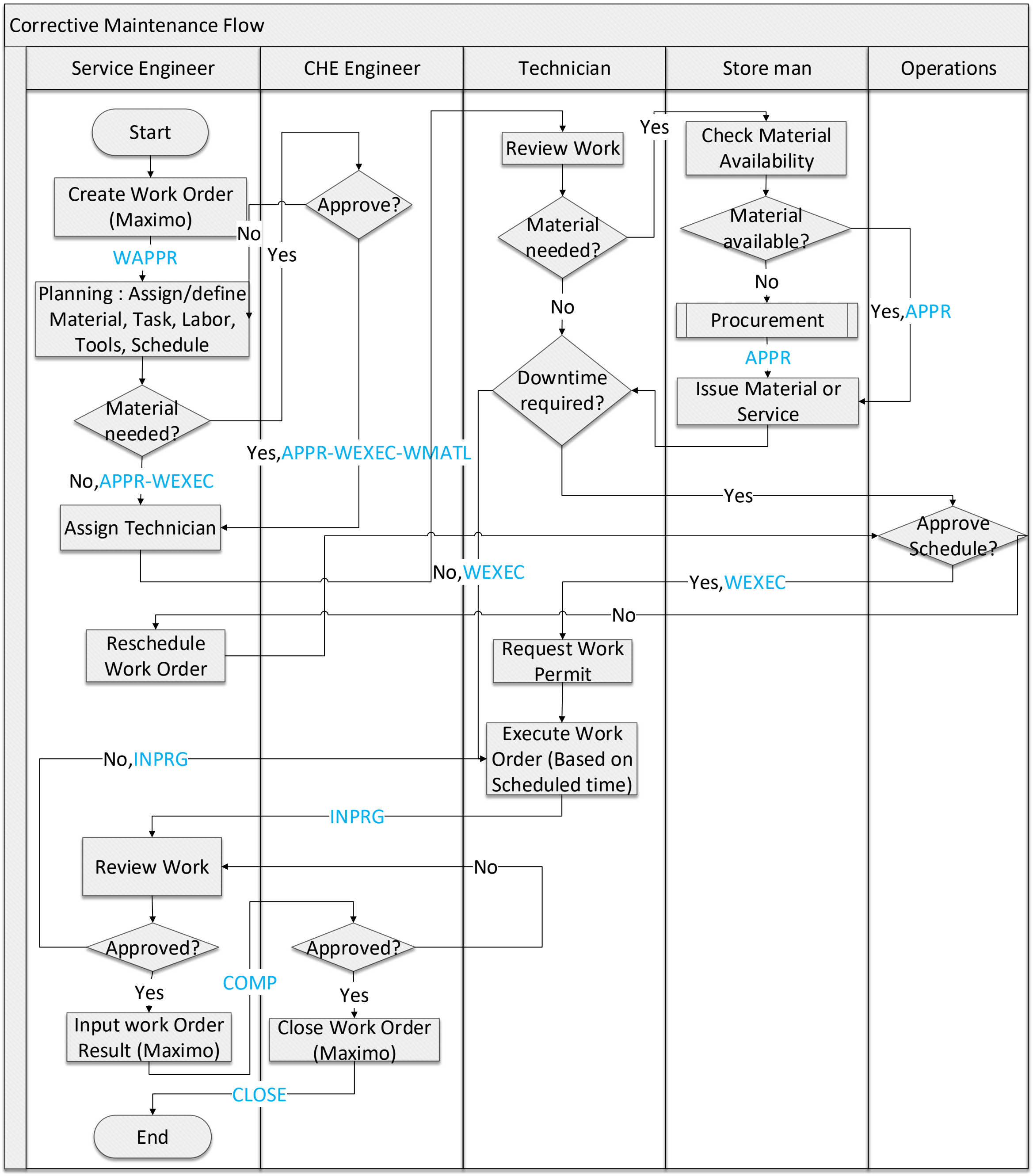
Corrective maintenance (CM) work is a maintenance work initiated based on failure happens on assets or locations. The purpose of this maintenance work is to return an asset or location into a satisfactory operating condition. CM work order created in **Work Order Tracking** application.

Planning should be held before executing a CM work. A job plan can be defined in a CM work order to automate the planning resources and tasks need to be performed, or creator can defined them manually. Approval from the CM work order is needed when there are materials required for the work.

Most CM work requires downtime of the assets to be performed, but not always. When downtime required, the work schedule need to be considered more carefully, to minimalize the impact on business process. If the schedule does not fit, a schedule revision should be done towards those CM work order.

5.2 CM Work Cycle

For the details of corrective maintenance process flow, please refers to below image.



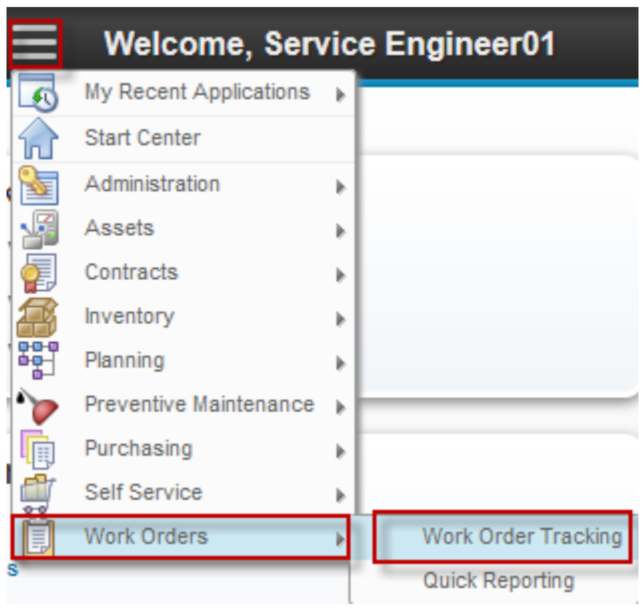
Below are detail scenario for Corrective Maintenance (CM) work order:

1. CM work order created from **Work Order Tracking** application by *Service Engineer*, based on prior inspection. The work order creator must set the *Work Type* attribute value into **CM**.
2. CM Work Order plans can be defined by selecting job plan that apply, or planned manually by work order creator.

3. When there are materials planned to perform this CM work, an approval assignment will be sent to *CHE Engineer* to issue those materials from the storeroom. If the materials are unavailable in the storeroom, procurement should be held to purchase them.
4. *Operation* approval is required related to CM work order schedule, if the work order requires downtime. *Service Engineer* should revise the schedule if it does not fit.
5. *Service Engineer* reviews the work result and input the result into EMS. A closing approval assignment will be sent to *CHE Engineer*, whether the work order can be closed or not.

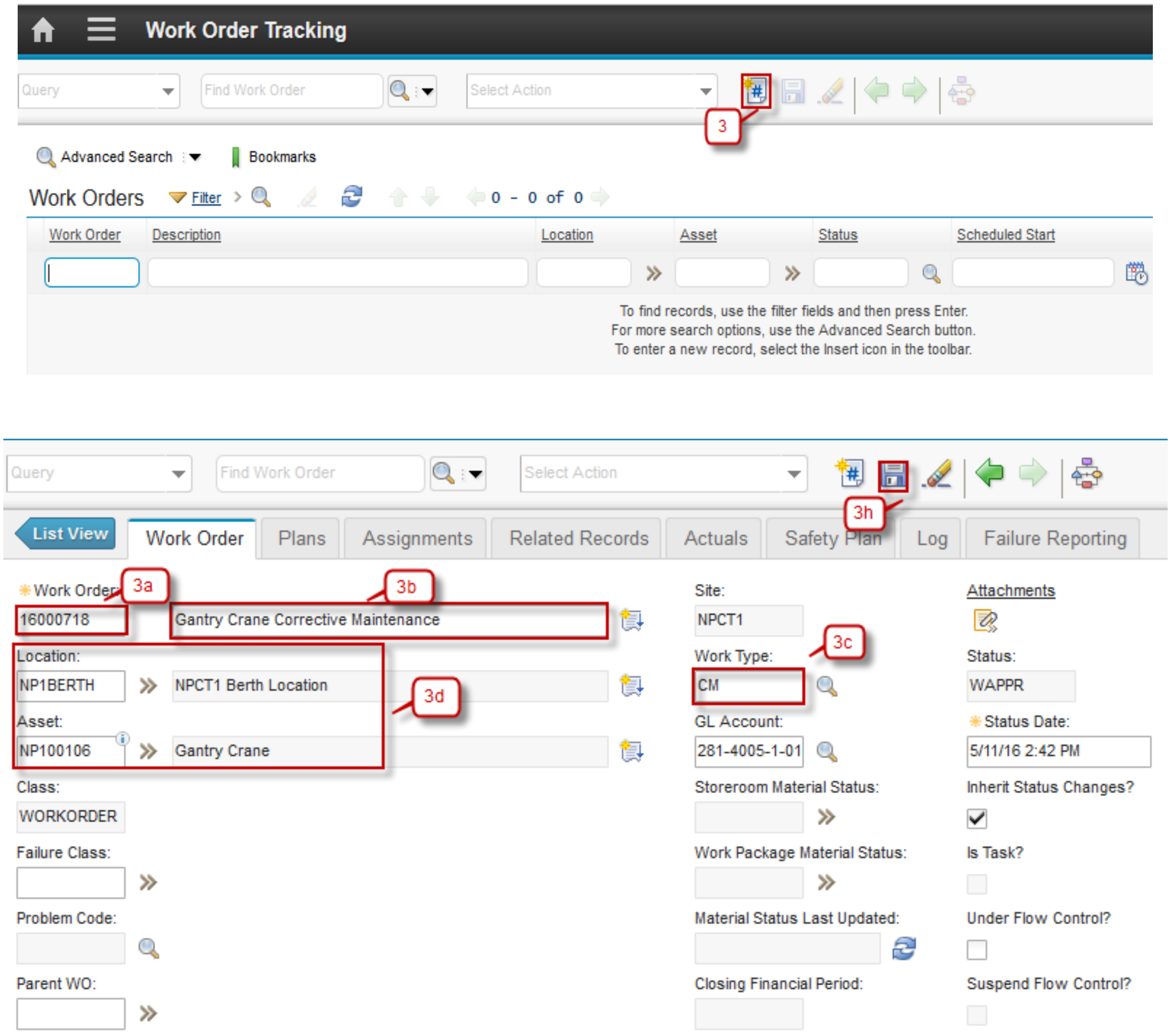
5.2.1 CM Work Creation & Submission

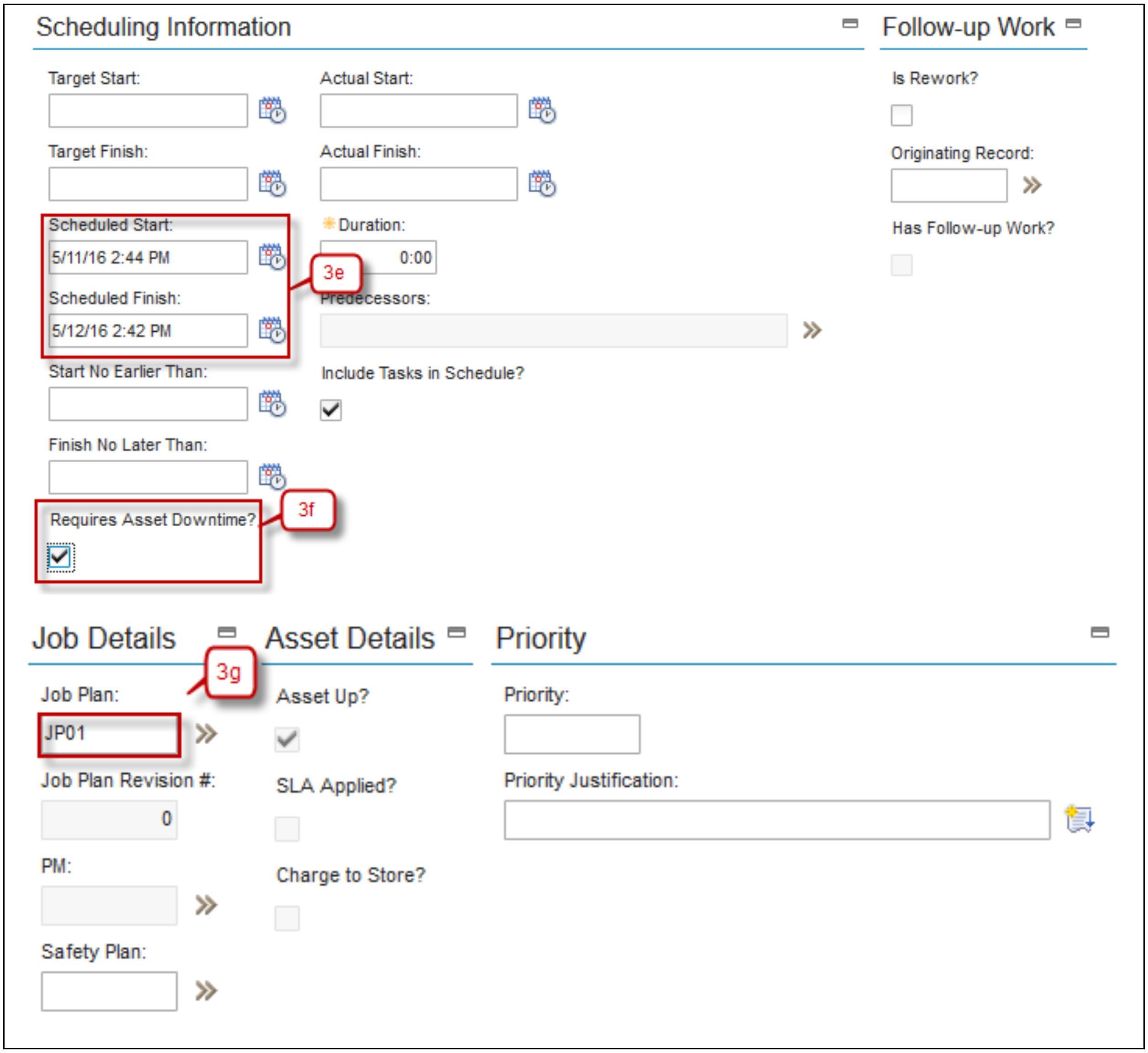
Below are step by step for CM Work Order creation & submission:

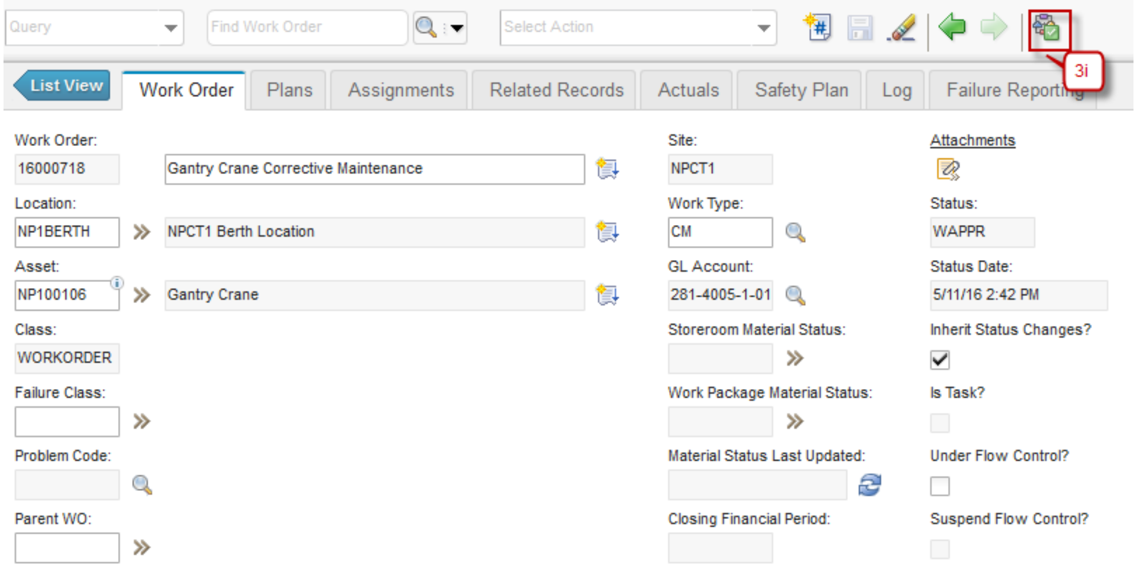
Step	Action
1. Login as Service Engineer. User name : se01, password : password.	
2. Go to Work Order Application by click **(GoTo) – Work Orders – Work Order Tracking**	
3. Click **New Work Order** button 3a. Work Order number is autonumber. 3b. Input the Work Order description. 3c. Select **Work Type 'CM' (Corrective Maintenance).** 3d. Select Asset (**mandatory field**). Location automatically filled in accordance with the location of assets. 3e. Input Schedule Start and Schedule Finish. 3f. Check the **Requires Asset Downtime?** 3g. Select the Job Plan.	

3h. Click **Save** button

3i. The **Route Workflow** button appearance change to , indicates that the workflow is started.




Scheduling Information
Follow-up Work
Target Start:
Actual Start:
Target Finish:
Actual Finish:
Scheduled Start:
5/11/16 2:44 PM
Duration:
0:00
3e
Scheduled Finish:
5/12/16 2:42 PM
Predecessors:
Start No Earlier Than:
Include Tasks in Schedule?
Finish No Later Than:
Requires Asset Downtime?
3f
Is Rework?
Originating Record:
Has Follow-up Work?
Job Details
Asset Details
Priority
3g
Job Plan:
JP01
Job Plan Revision #:
0
PM:
Safety Plan:
Asset Up?
SLA Applied?
Charge to Store?
Priority:
Priority Justification:



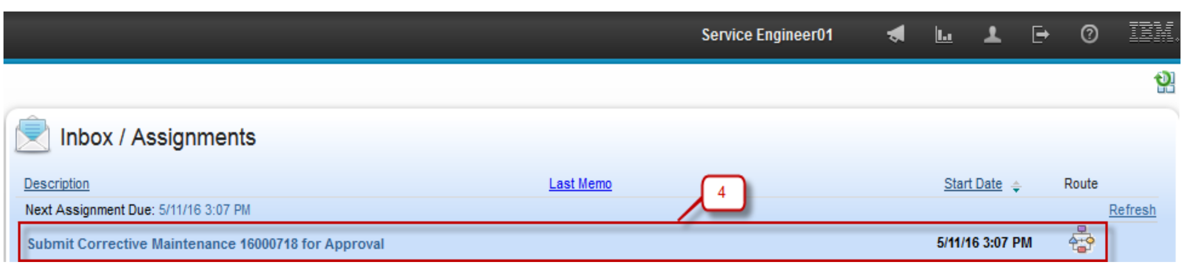
4. Go to Start Center Service Engineer, select work order assignment that displayed on **Inbox/Assignment** column.

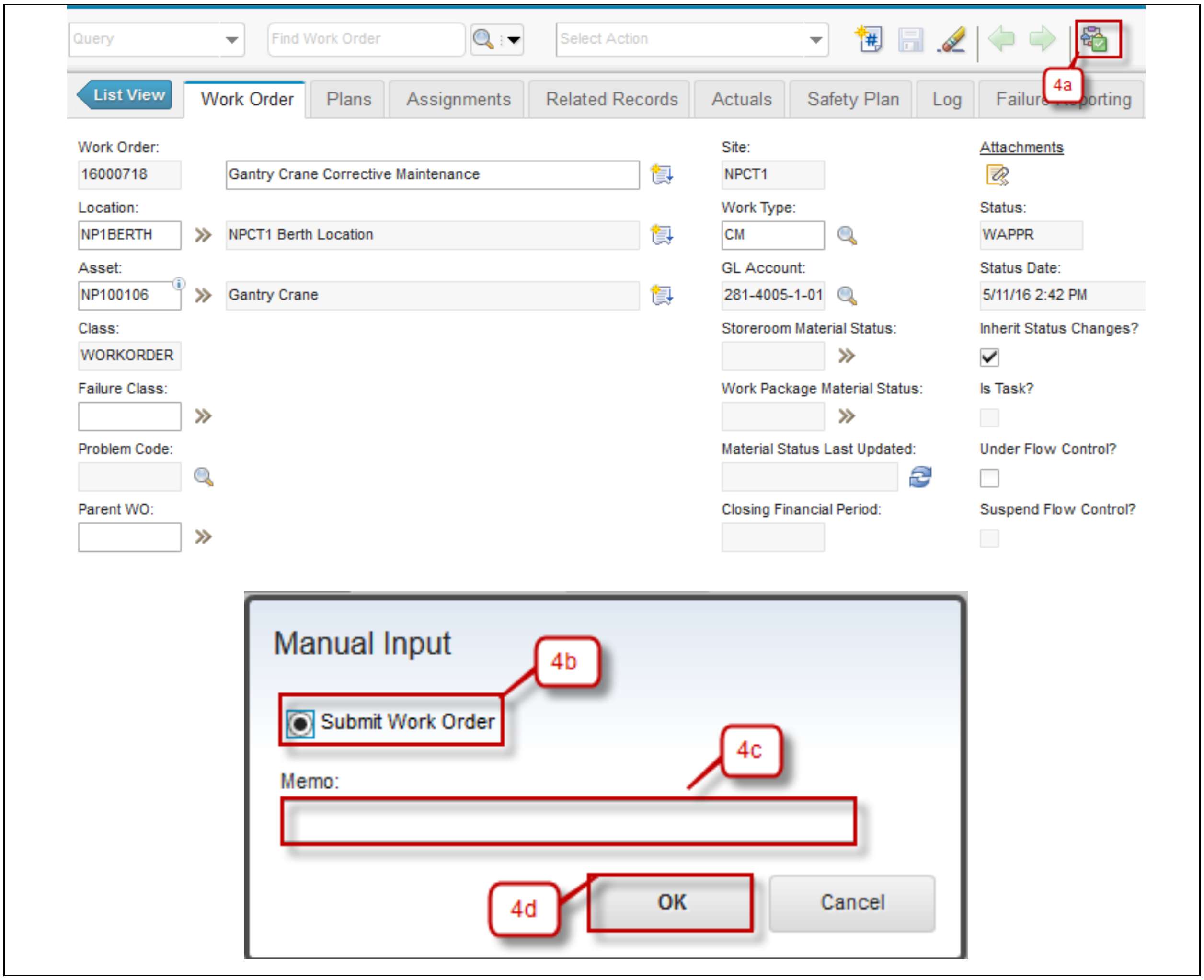
4a. Click  (Route Workflow) button and dialog box to submit Work Order will appears.

4b. Choose **Submit Work Order.**

4c. Input Memo (if needed).

4d. Click **OK** button.





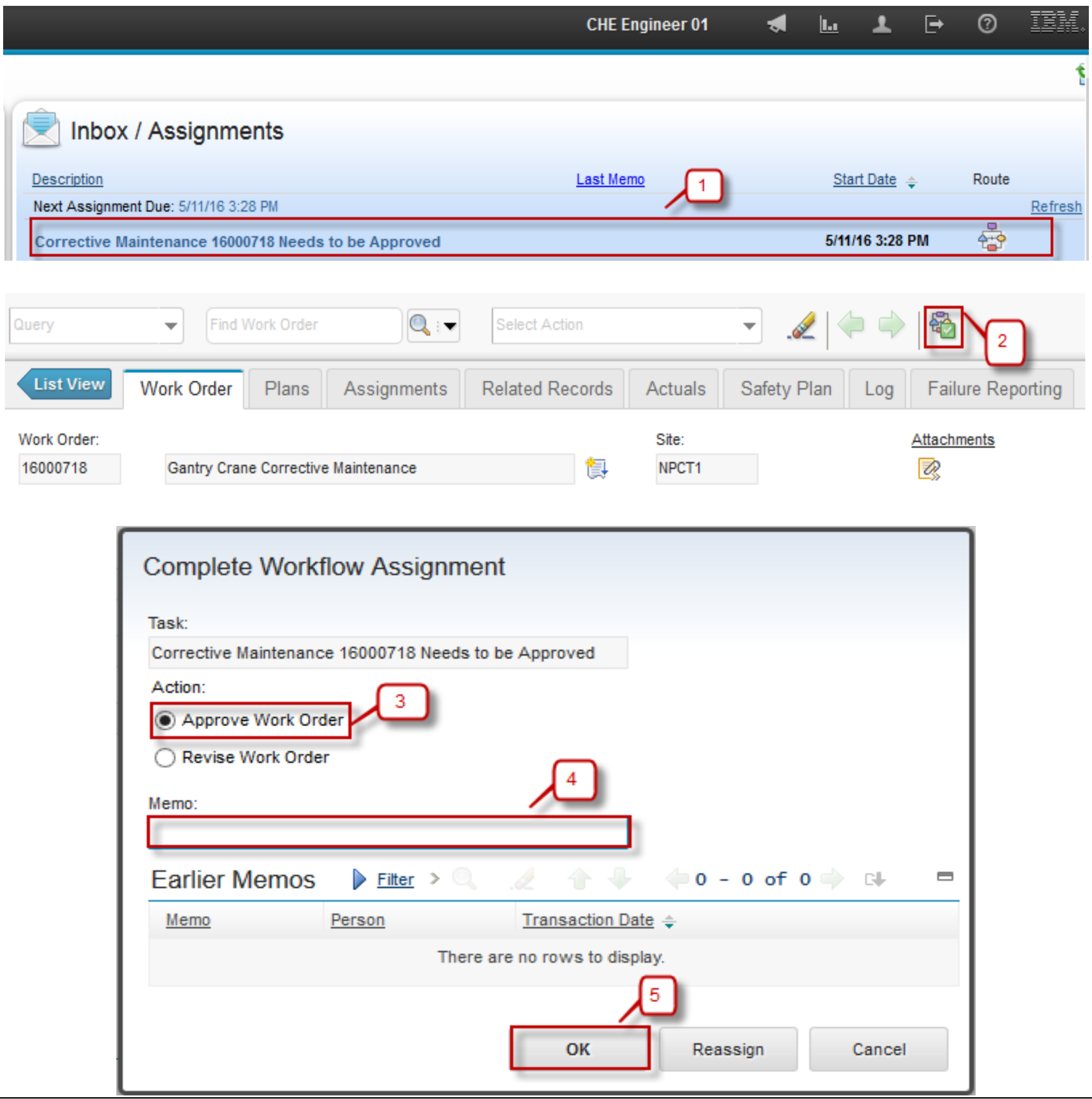
5.2.2 CM Work Plan Approval Process by CHE Engineer

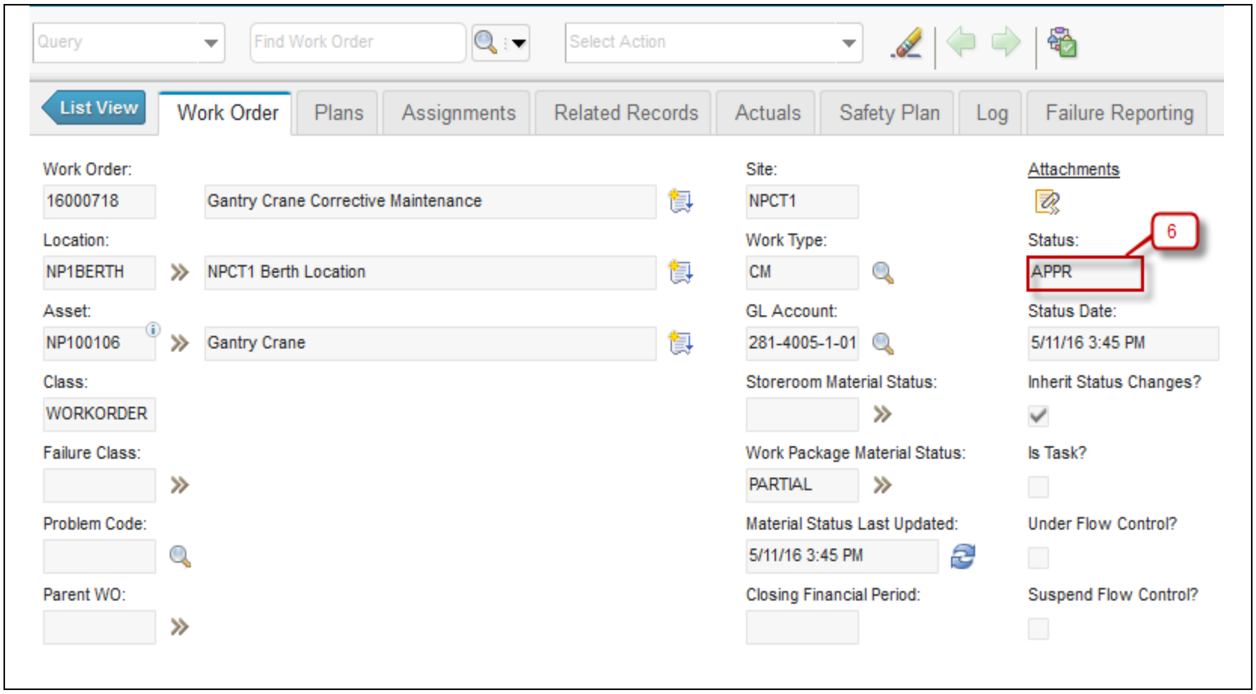
CHE Engineer approval is required when material is planned for the work order. Below are steps to approve CM work plan by *CHE Engineer*. For rejection, revision request, and revision will not be discussed. They can refer to rejection, revision request, and revision of **SM Work Cycle** in step point **1.2.5 SM Request Revision Work Order** and **1.2.6 SM Revise and Re-submit Work Order** training material.

As a note, request revision for CM work plan by *CHE Engineer* will direct the assignment back to *Service Engineer*. After revision is done, assignment will be directed back to *CHE Engineer* for plan approval.

Step	Action

5. Login as CHE Engineer. Username : che01, password: password. On Start Center CHE Engineer an assignment to approve created Work Order will appears in **Inbox/Assignment** portlet, click Description of the assignment to go to related Work Order.
6. Then click (route workflow) and dialog box Complete Workflow Assignment will appears.
7. Choose **Approve Work Order.**
8. Input Memo (if needed).
9. Click **OK.**
10. Status change to '**APPR**' (**Approve**).

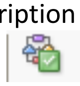


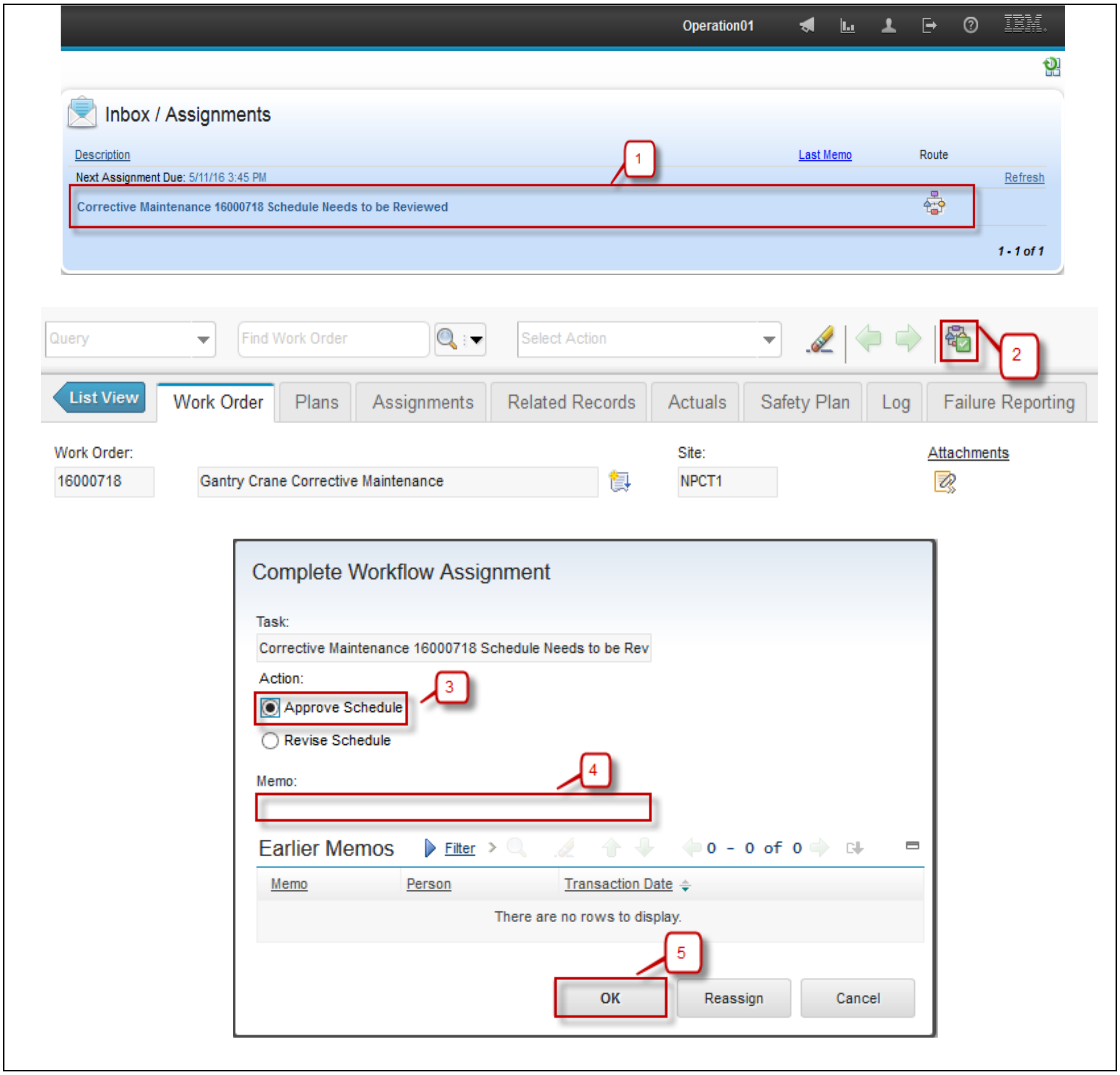


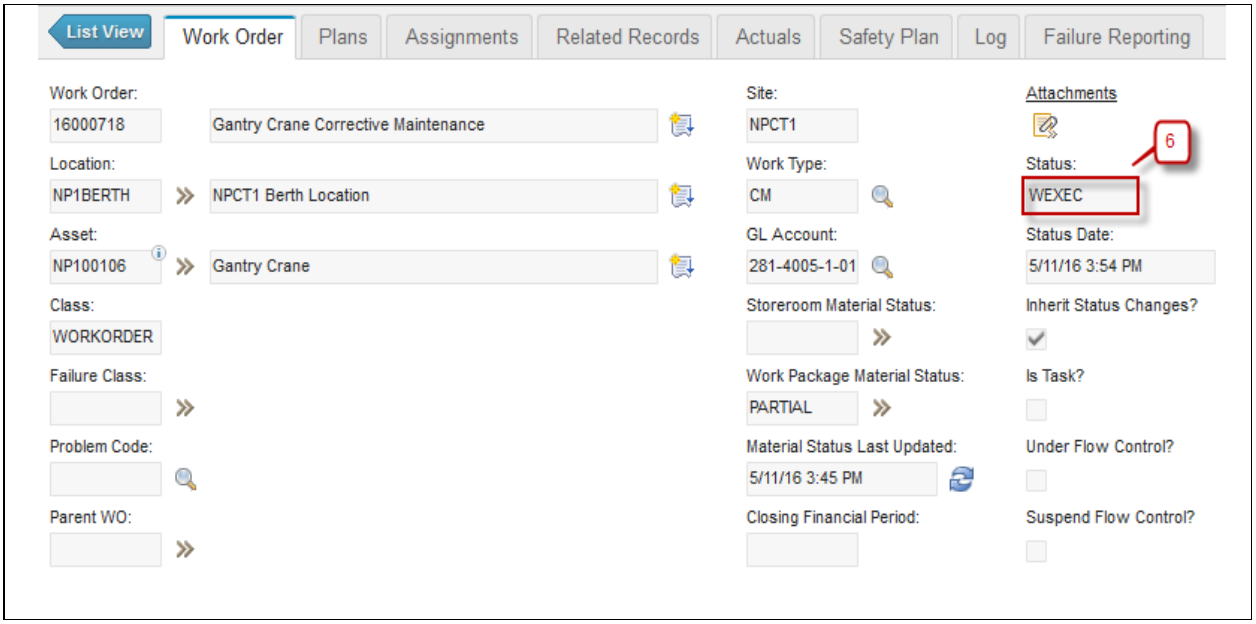
5.2.3 CM Work Schedule Approval by Operation

After CM work plan is approved, if asset downtime is required to execute the work, the work schedule needs to be approved by *Operation.* Below are steps to approve a CM work schedule by *Operation*. For rejection, revision request, and revision will not be discussed. They can refer to rejection, revision request, and revision of **SM Work Cycle** in step point **1.2.5 SM Request Revision Work Order** and **1.2.6 SM Revise and Re-submit Work Order** training material.

As a note, request revision for CM work schedule by *Operation* will direct the assignment back to *Service Engineer*. After revision is done, assignment will be directed back to *Operation* for schedule approval.

Step	Action
1.	Login as Operation. Username : ops01, password : password. On Start Center Operation an assignment to review approved Work Order schedule will appears in **Inbox/Assignment** portlet, click Description of the assignment to go to related Work Order.
2.	Then click  (route workflow) and dialog box Complete Workflow Assignment will appears.
3.	Choose **Approve Schedule.**
4.	Input Memo (if needed).
5.	Then click **OK.**
6.	Status change to '**WEXEC**' (Waiting for Execution)

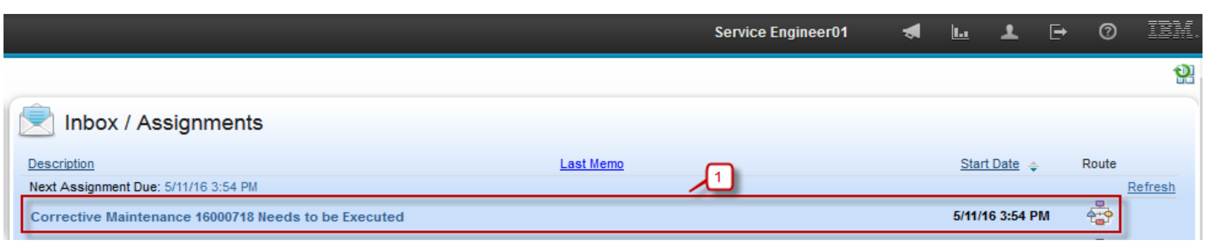

Operation01
IBM.
Inbox / Assignments
Description
Last Memo
Route
Next Assignment Due: 5/11/16 3:45 PM
Refresh
Corrective Maintenance 16000718 Schedule Needs to be Reviewed
1 - 1 of 1
1
Query
Find Work Order
Select Action
2
List View
Work Order
Plans
Assignments
Related Records
Actuals
Safety Plan
Log
Failure Reporting
Work Order:
16000718
Gantry Crane Corrective Maintenance
Site:
NPCT1
Attachments
Complete Workflow Assignment
Task:
Corrective Maintenance 16000718 Schedule Needs to be Rev
Action:
Approve Schedule
3
Revise Schedule
Memo:
4
Earlier Memos
Filter
0 - 0 of 0
Memo
Person
Transaction Date
There are no rows to display.
5
OK
Reassign
Cancel

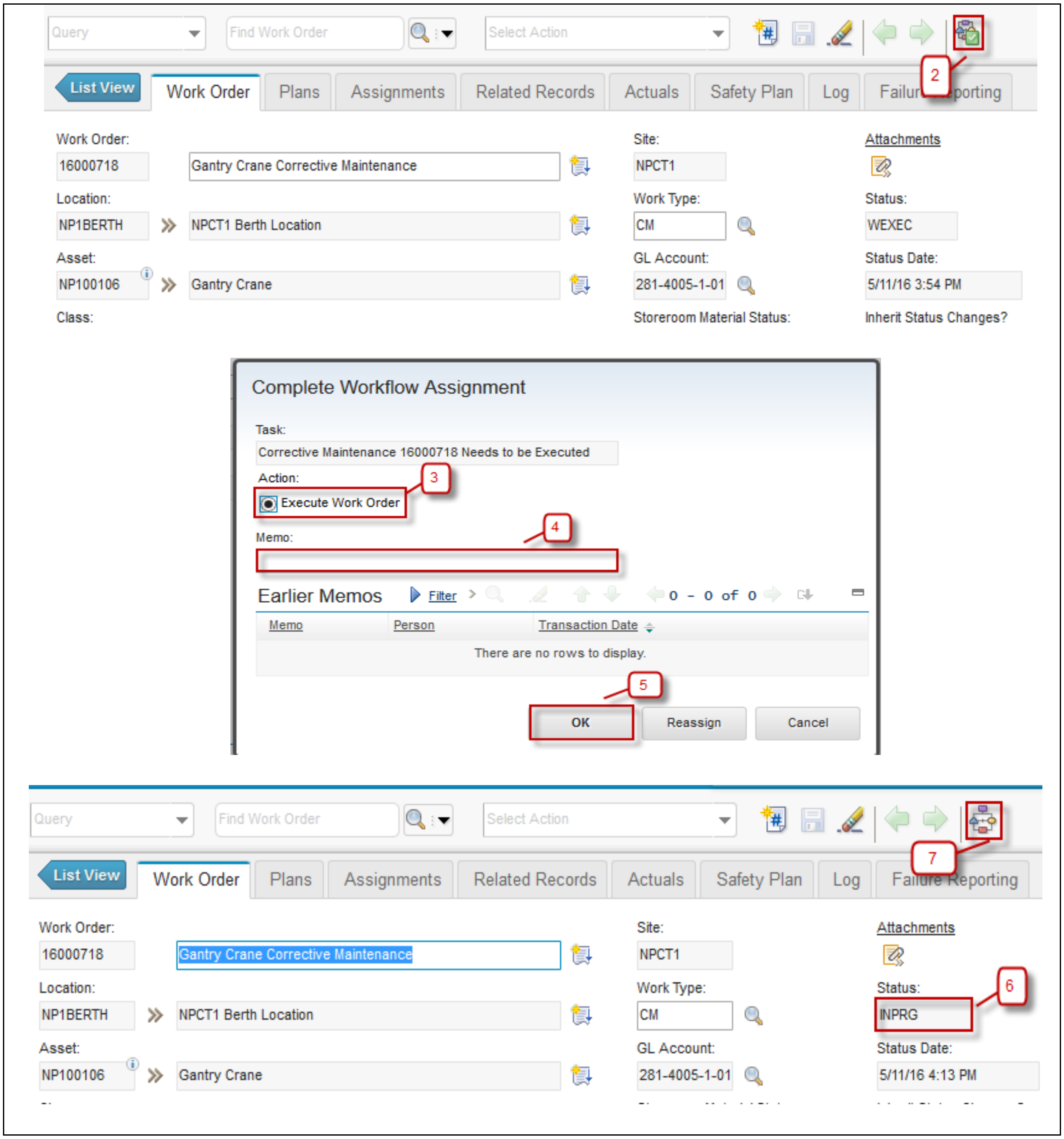


5.2.4 CM Work Execution

To initiate CM work, below are step by step needs to be done by *Service Engineer* on CM work order:

Step	Action
1.	Login as Service Engineer. Username : se01, password : password. On Start Center Service Engineer an assignment to execute Work Order will appears in **Inbox/Assignment** portlet, click Description of the assignment to go to related Work Order.
2.	Then click  (route workflow) and dialog box Complete Workflow Assignment will appears.
3.	Choose **Execute Work Order.**
4.	Input Memo (if needed).
5.	Then click **OK.**
6.	Status change to '**INPRG**' (In Progress).
7.	The **Route Workflow** button appearance returned back into , indicates that the workflow is finished.





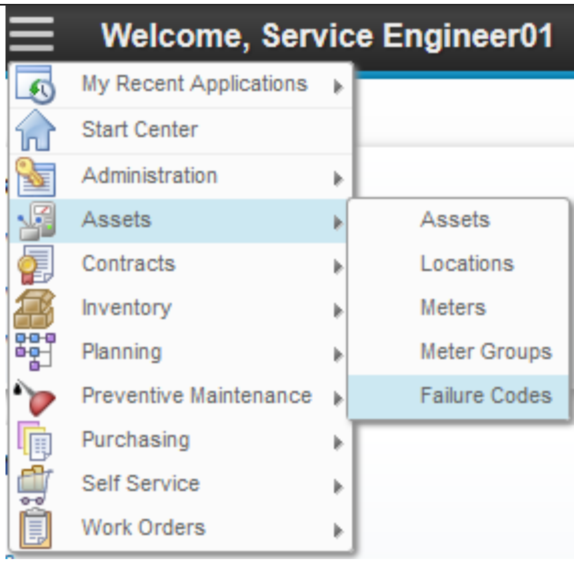
5.2.5 Failure Codes Application

For every corrective work, whether it is CM or SR work (SR work will be discussed next), cause of failure and the solution must be defined before it is being closed. This regulation stipulated that companies have a source of knowledge about the problems that may occur and how to overcome them. A failure codes should be defined for each work order, containing:

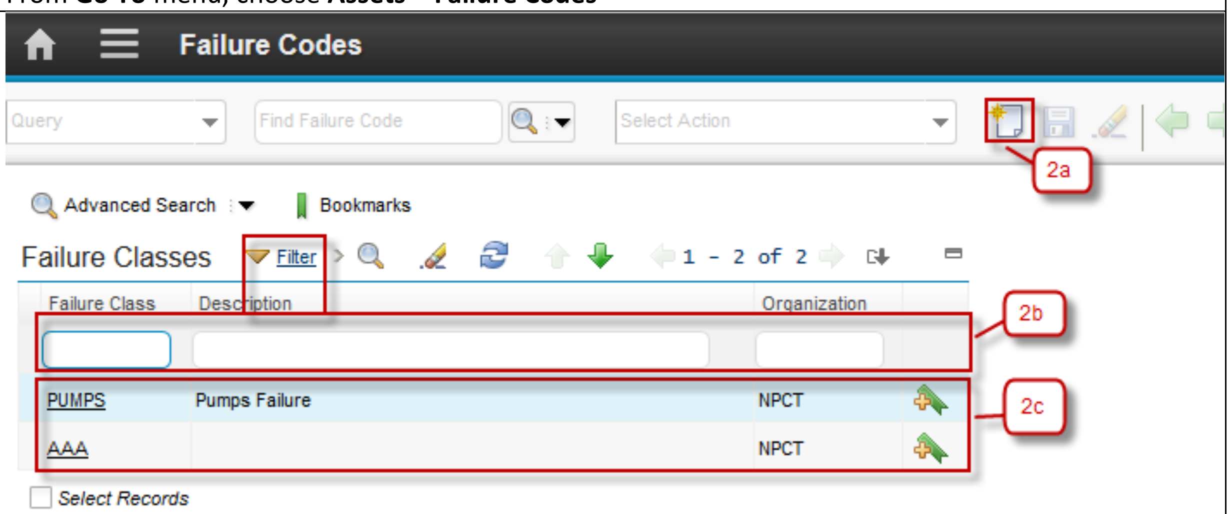
- *Problem.* Define the problem that occurs.
- *Cause*. Define the cause of the problem.
- *Remedy*. Define the solution of the problem.

A failure codes must be defined by *Service Engineer* for CM/SR work order before the work order is submitted for closing. Failure codes can be chosen from available list, defined in **Failure Codes** application. Below are details about Failure Codes applications accessibility:

<table>
<tr>
<td>1</td>
<td>

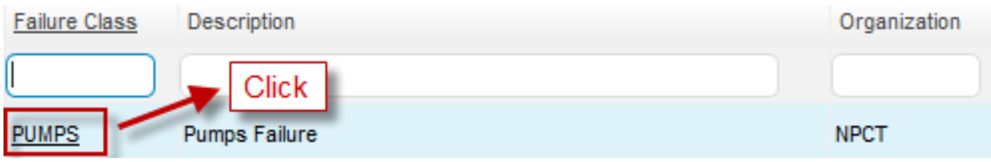


From **Go To** menu, choose **Assets – Failure Codes**
</td>
</tr>
<tr>
<td>2</td>
<td>

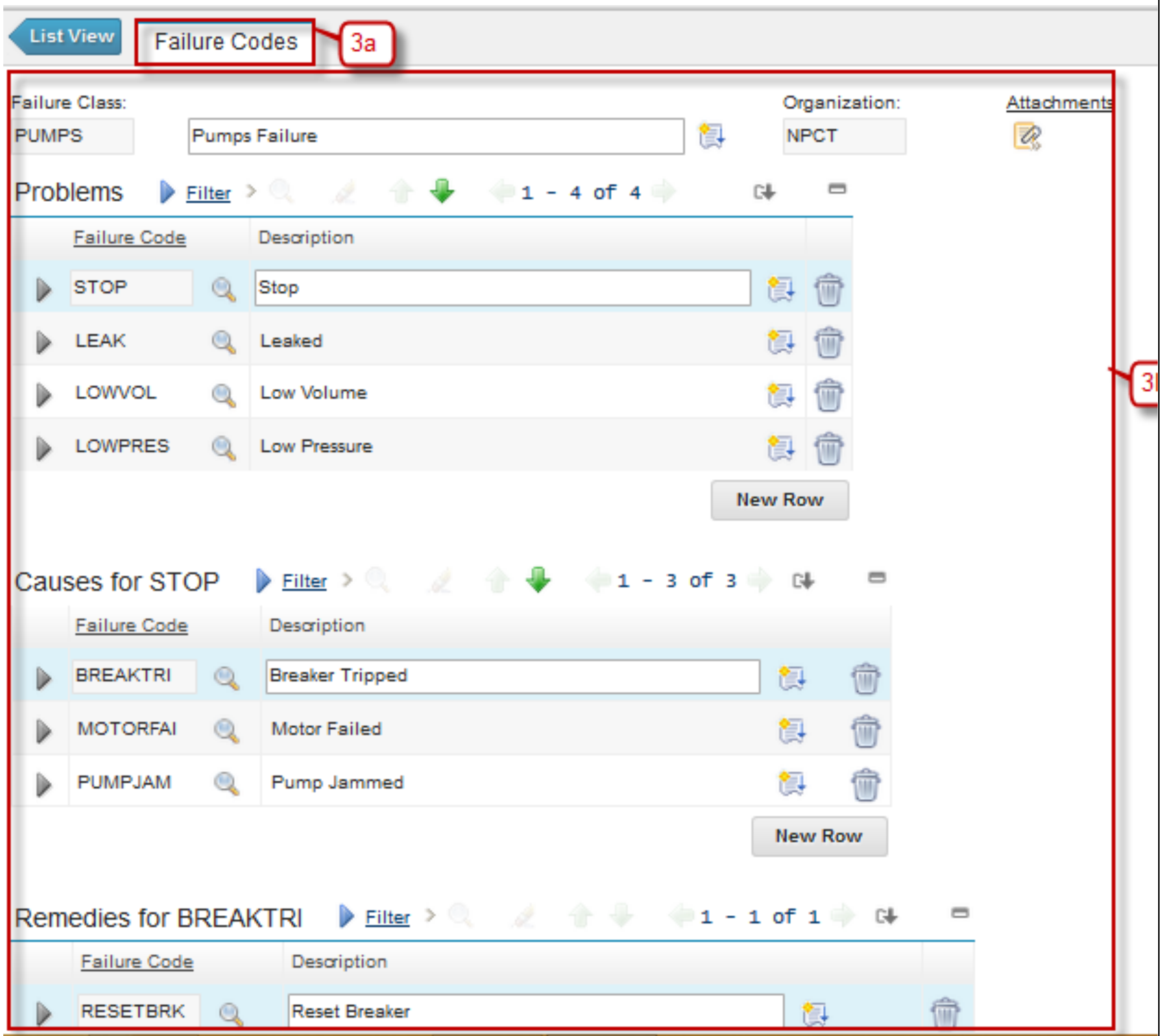


Header Bar will give information that user is in the Failure Codes application.
2.a **New Failure Code** button, to add new Failure Code
2.b **Filter** Failure Classes by **Failure Class, Description, and Organization.**
</td>
</tr>
</table>

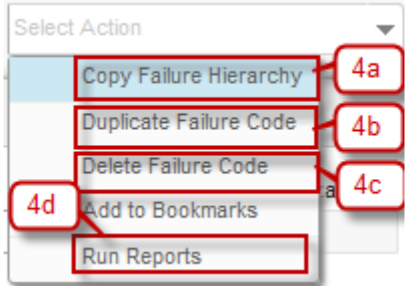
<table>
<tr><td></td><td>2.c List of all failure class based on filtering result.</td></tr>
<tr><td>3</td><td>Choose one Failure Class, to show the details of the Failure Class.



Will show display the details, see the figure below.

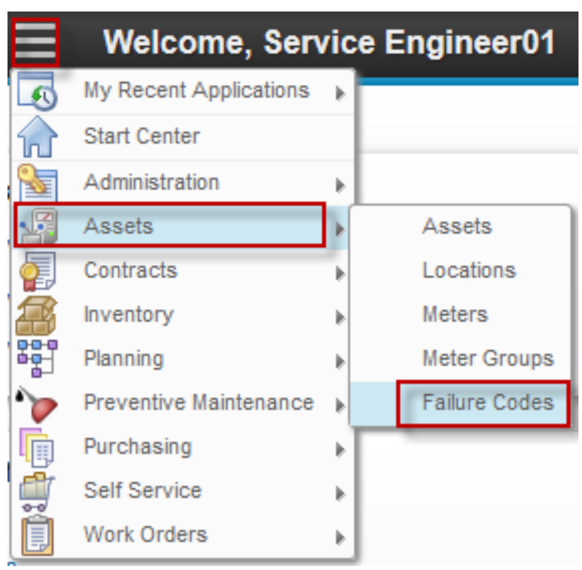


3.a **Failure Codes** tab, to view and update details of Failure Code
3.b Detail information of the Problem, Causes, and Remedies.</td></tr>
<tr><td>4</td><td>User can perform some action related to the Failure Code from Select Action menu (on top of the form).</td></tr>
</table>

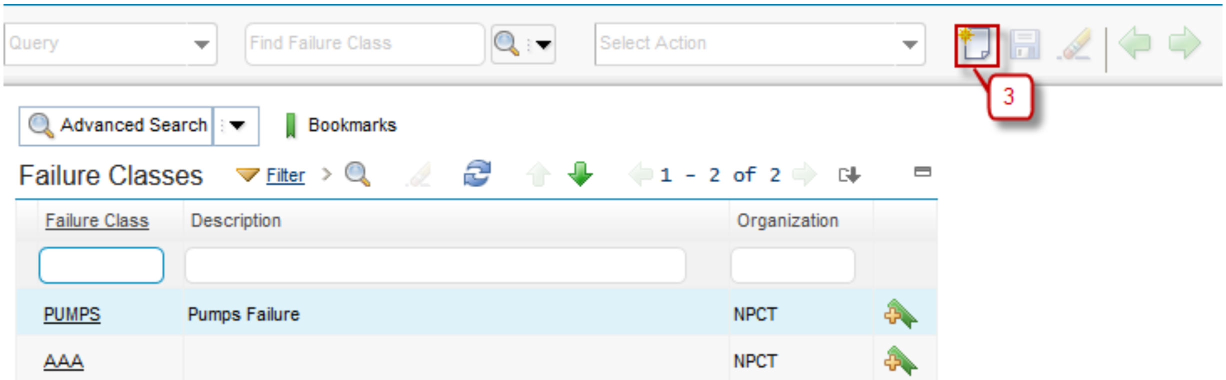
	 4.a **Copy Failure Hierarchy** action to copy failure hierarchy of every problem. 4.b **Duplicate Failure Code,** to create new failure code by duplicating most value of the current failure code. 4.c **Delete Failure Code,** to delete failure code. 4.d **Run Reports,** to run reports of failure code.

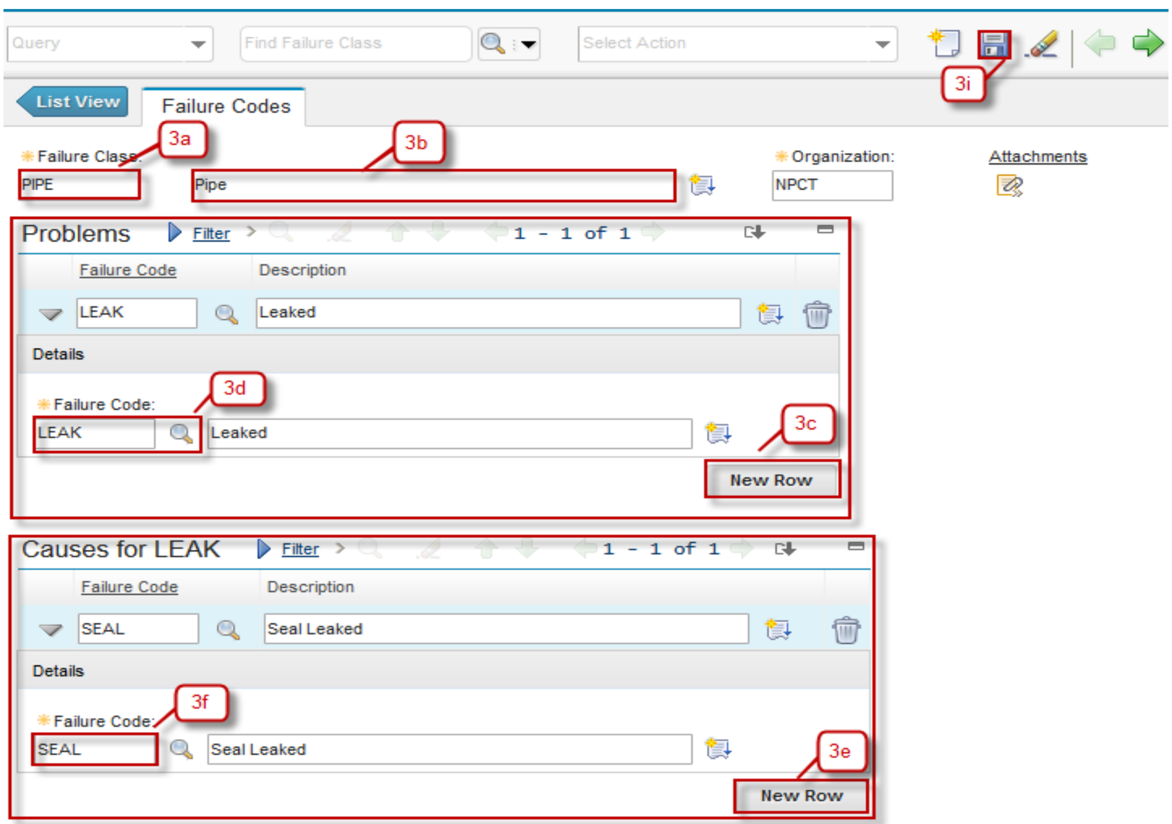
5.2.5.1 Create Failure Codes

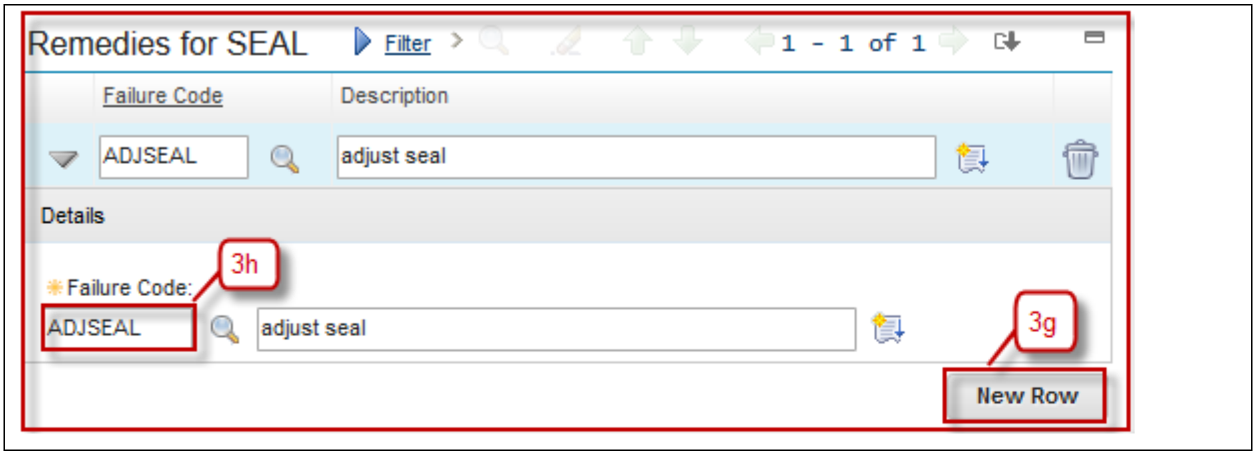
When new failure code need to be defined, or new problem, cause, remedy need to be added in a failure code, *Service Engineer* can perform that in **Failure Codes** application, by following below steps:

Step	Action
1. Login as Service Engineer. User name : se01, password : password.	
2. Go to Work Order Application by click **(GoTo) – Assets – Failure Codes**	
3. Click **New Work Order** button 3.a Input the failure class name on **Failure Class** column 3.b Input description of that failure class. 3.c Click **New Row** on **Problems** list.	

3.d Select Failure Code of Problem.
3.e Click **New Row** on **Causes** list.
3.f Select Failure Code of Causes.
3.g Click **New Row** on **Remedies** list.
3.h Select Failure Code of Remedies.
3.i Click **Save** button.





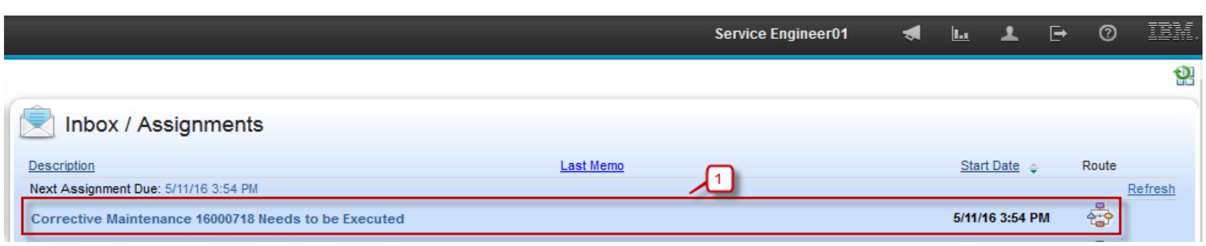


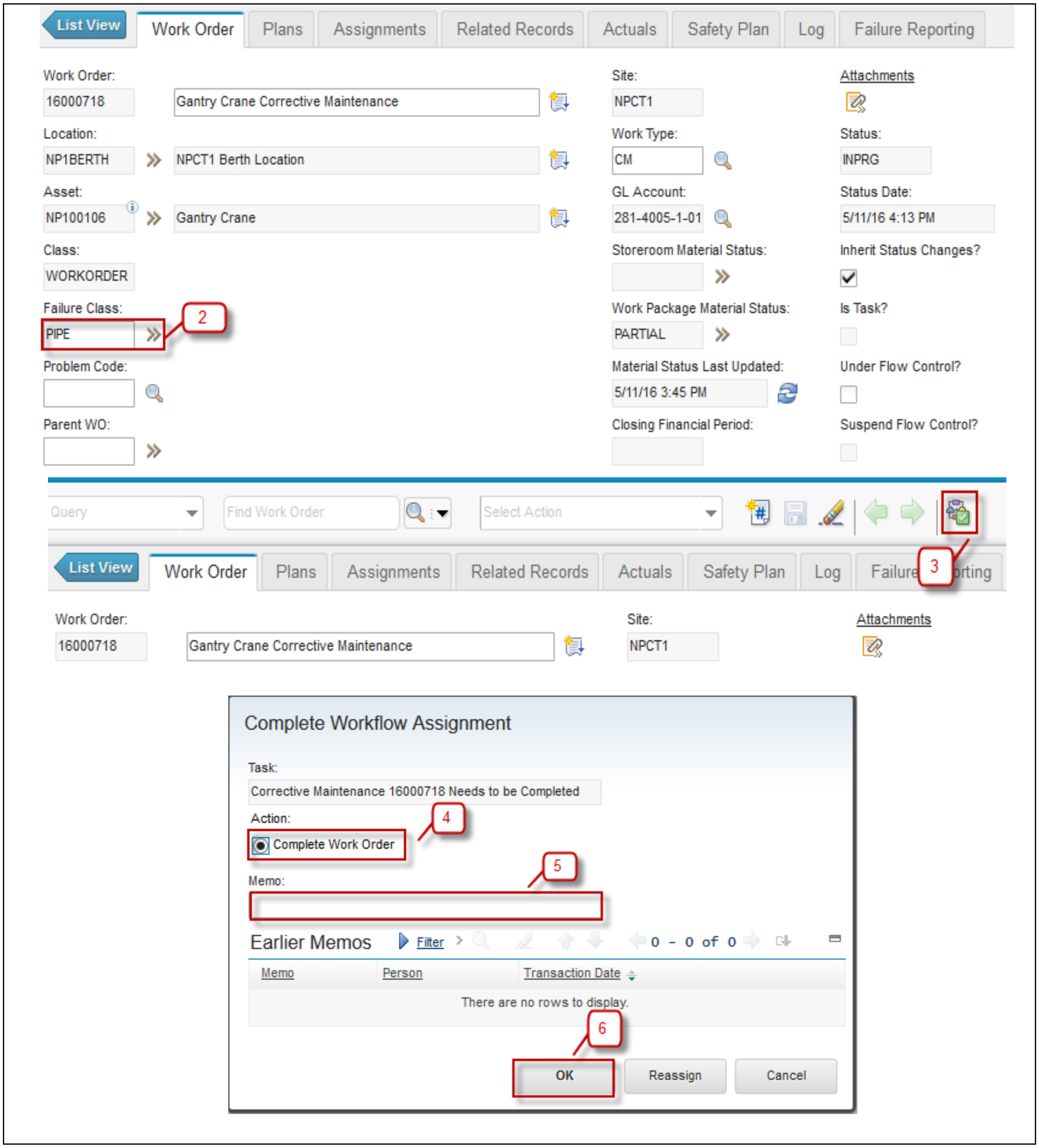
5.2.6 CM Work Completion

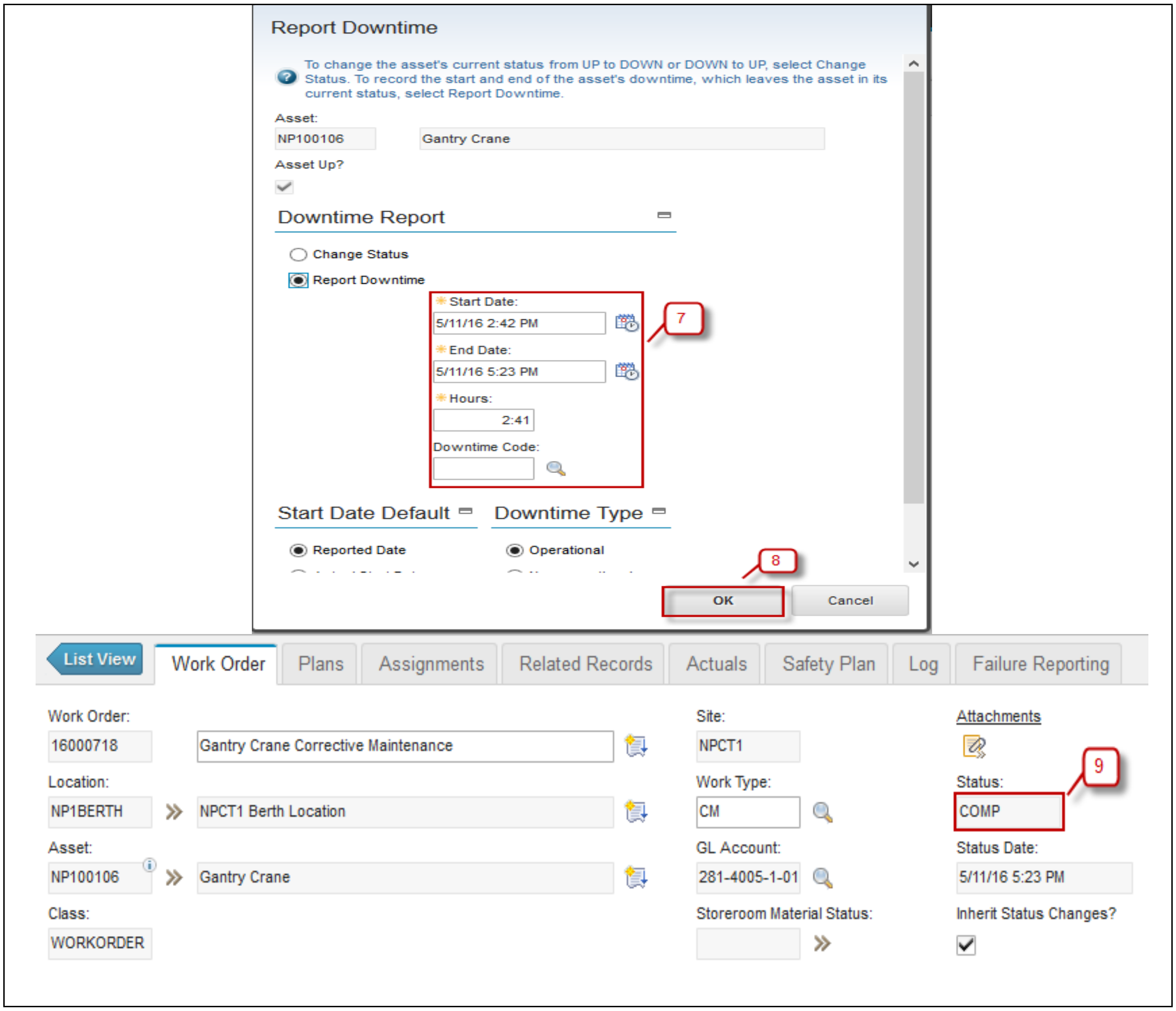
When CM work is finished, *Service Engineer* should input the work result in the CM work order (including failure codes). After that, the work order must be submitted for closing. Below are step by step to complete SM Work Order:

Step	Action

1. Login as Service Engineer. Username : se01, password : password. Choose one of related work order which has "**INPRG**" status and work type '**CM**' (**Corrective Maintenance**).
2. Select the Failure Code for this Work Order, click  button to select failure code and then click **Save** button.
3. Then click  (route workflow) and the **Route Workflow** button appearance returned back into , indicates that the workflow is started.
4. Choose **Complete Work Order.**
5. Input Memo (if needed).
6. Then click **OK.**
7. Dialog box **Report Downtime** will be appeared and field Start Date and End Date automatically filled. Value of Hours is interval between Start Date and End Date.
8. Click **OK**
9. Status change to '**COMP** (Complete).




List View
Work Order
Plans
Assignments
Related Records
Actuals
Safety Plan
Log
Failure Reporting
Work Order:
16000718
Gantry Crane Corrective Maintenance
Site:
NPCT1
Attachments
Location:
NP1BERTH
NPCT1 Berth Location
Work Type:
CM
Status:
INPRG
Asset:
NP100106
Gantry Crane
GL Account:
281-4005-1-01
Status Date:
5/11/16 4:13 PM
Class:
WORKORDER
Storeroom Material Status:
Inherit Status Changes?
Failure Class:
PIPE
2
Work Package Material Status:
PARTIAL
Is Task?
Problem Code:
Material Status Last Updated:
5/11/16 3:45 PM
Under Flow Control?
Parent WO:
Closing Financial Period:
Suspend Flow Control?
Query
Find Work Order
Select Action
3
Complete Workflow Assignment
Task:
Corrective Maintenance 16000718 Needs to be Completed
Action:
Complete Work Order
4
5
Memo:
Earlier Memos
Filter
0 - 0 of 0
Memo
Person
Transaction Date
There are no rows to display.
6
OK
Reassign
Cancel



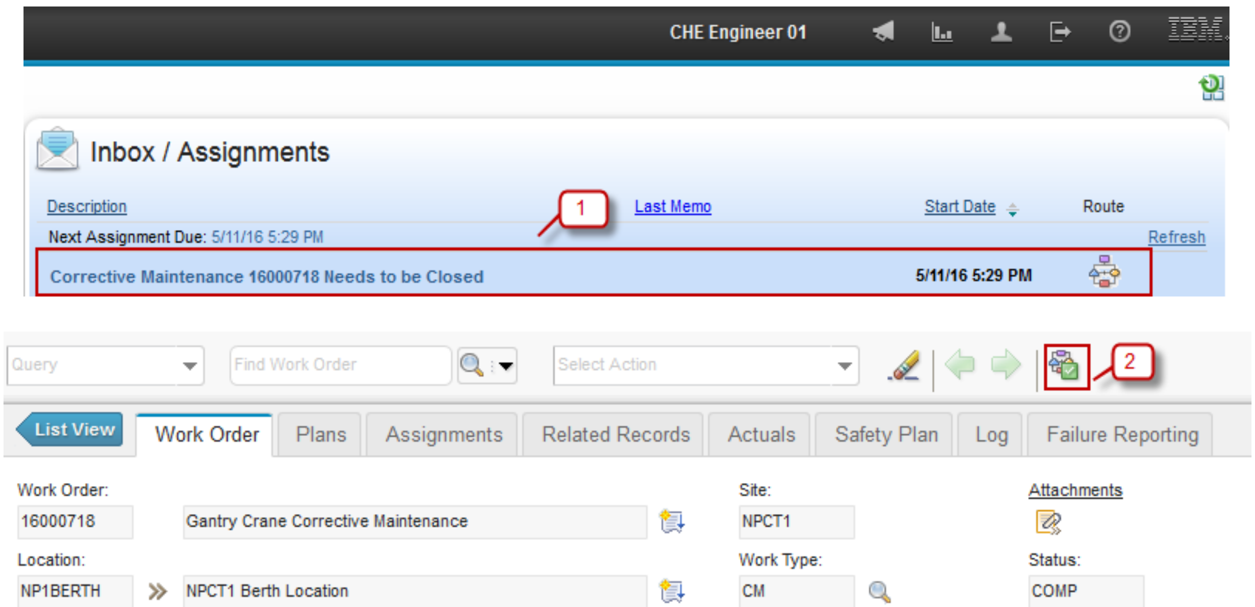
5.2.7 CM Work Closing Approval by CHE Engineer

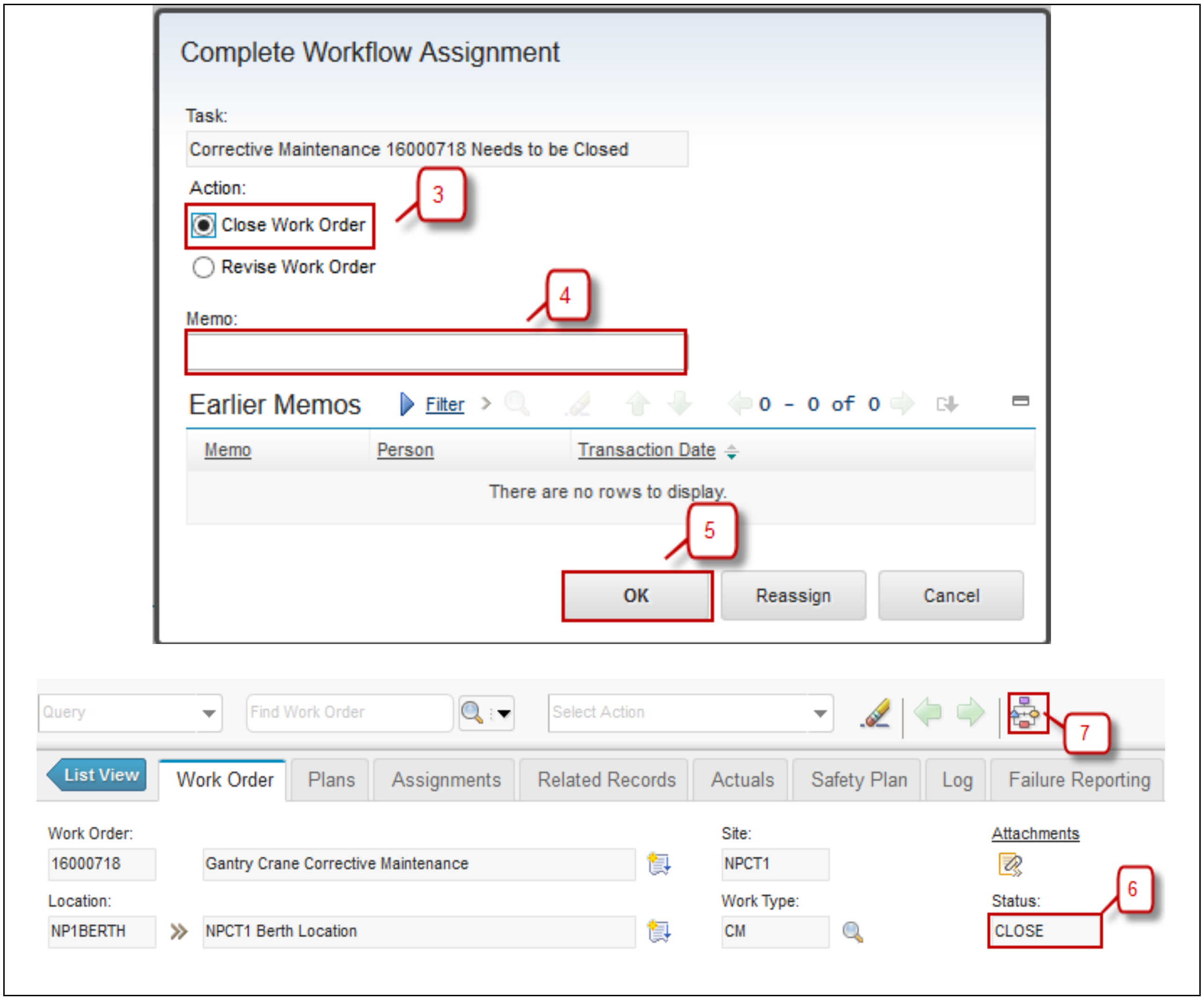
Below are steps to approve a CM work closing by *CHE Engineer*. For rejection, revision request, and revision will not be discussed. They can refer to rejection, revision request, and revision of **SM Work Cycle** in step point **1.2.5 SM Request Revision Work Order** and **1.2.6 SM Revise and Re-submit Work Order** training material.

As a note, request revision for CM work closing will direct the assignment back to *Service Engineer*. After revision is done, assignment will be directed back to *CHE Engineer* for closing approval.

Step	Action

1. Login as CHE Engineer. Username : che01, password: password. On Start Center Operation an assignment to close the Work Order will appears in **Inbox/Assignment** portlet, click Description of the assignment to go to related Work Order
2. Then click (route workflow) and dialog box Complete Workflow Assignment will appears.
3. Choose **Close Work Order**
4. Input Memo (if needed)
5. Then click **OK**
6. Status changed to '**CLOSE**' (Close).
7. The **Route Workflow** button appearance returned back into , indicates that the workflow is finished.




Complete Workflow Assignment
Task:
Corrective Maintenance 16000718 Needs to be Closed
Action:
Close Work Order
Revise Work Order
Memo:
Earlier Memos
Filter
0 - 0 of 0
Memo
Person
Transaction Date
There are no rows to display.
OK
Reassign
Cancel
3
4
5
Query
Find Work Order
Select Action
7
List View
Work Order
Plans
Assignments
Related Records
Actuals
Safety Plan
Log
Failure Reporting
Work Order:
16000718
Gantry Crane Corrective Maintenance
Site:
NPCT1
Attachments
Location:
NP1BERTH
NPCT1 Berth Location
Work Type:
CM
Status:
CLOSE
6

5.3 Exercises

6 Service Recovery (SR) Work

6.1 SR Work Overview

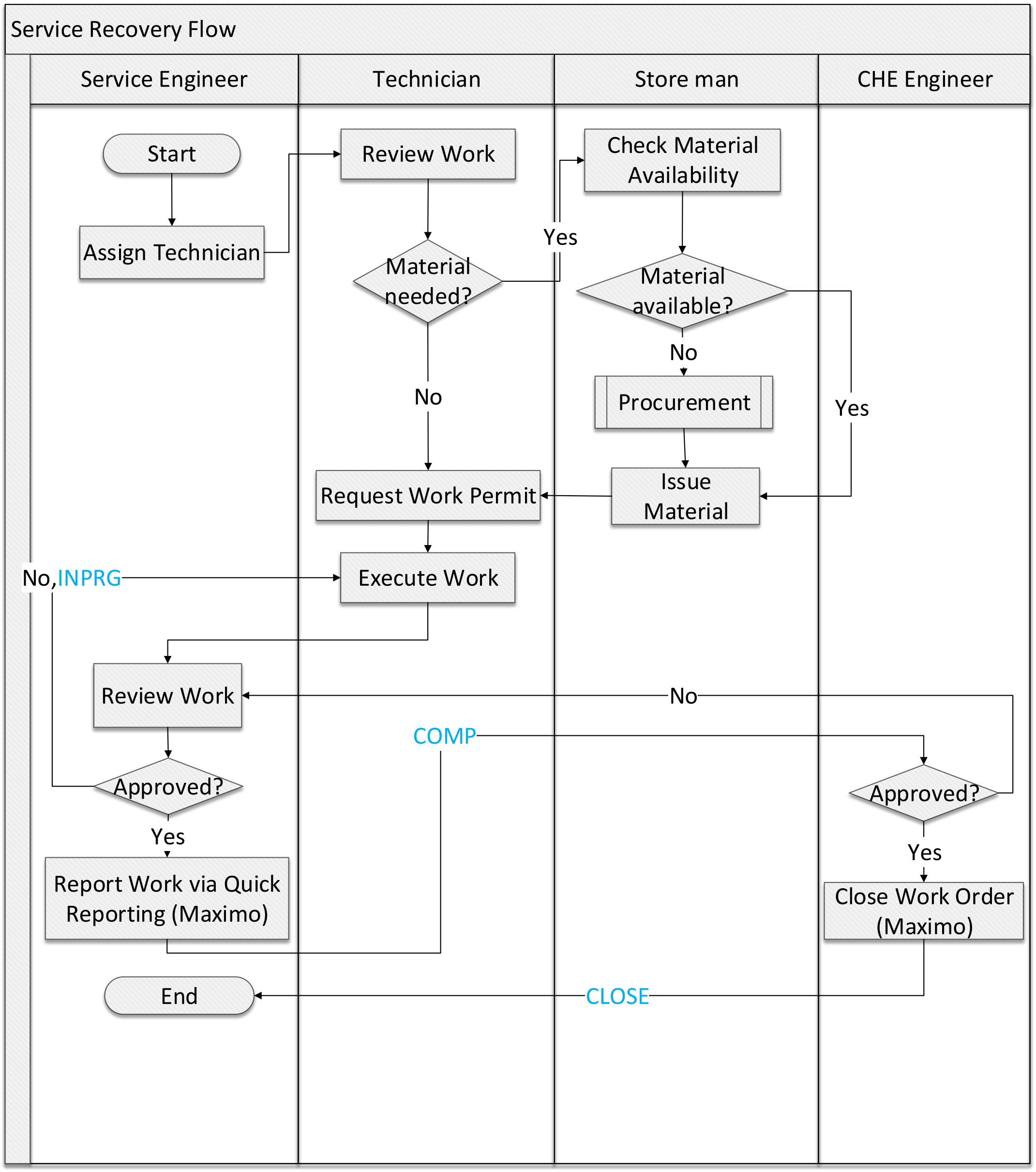
Service Recovery (SR) work is a corrective maintenance work with high level of urgency that makes them need to be executed immediately. Because of the high level of urgency, work execution can be done without prior work order. Material usage for this work does not require any prior approval too. After the work finished, report about this work can be registered as SR work order in **Quick Reporting** application.

Information that should be included in a SR work order is as follows:

- Tasks that have been performed
- Any materials or tools used.
- Labor resources used.
- Failure codes containing at least 1 problem, causes, and remedy.

6.2 SR Work Cycle

Below figure show how service recovery will be managed in EMS.



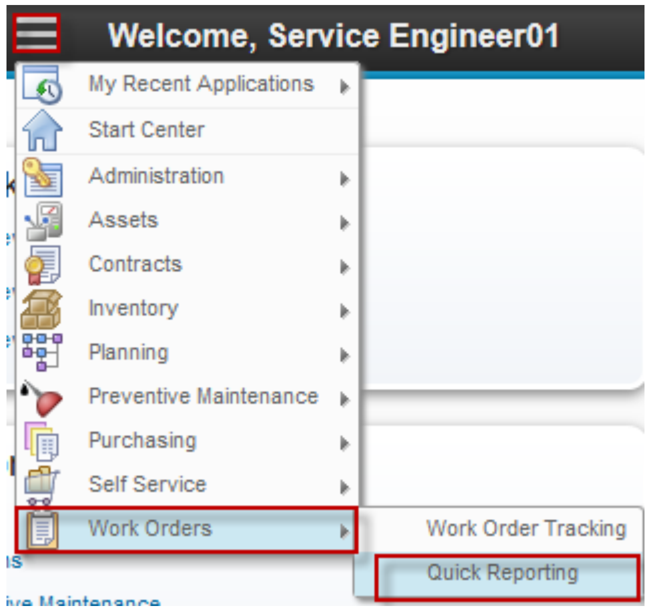
Below are detail scenario for Service Recovery (SR) work order:

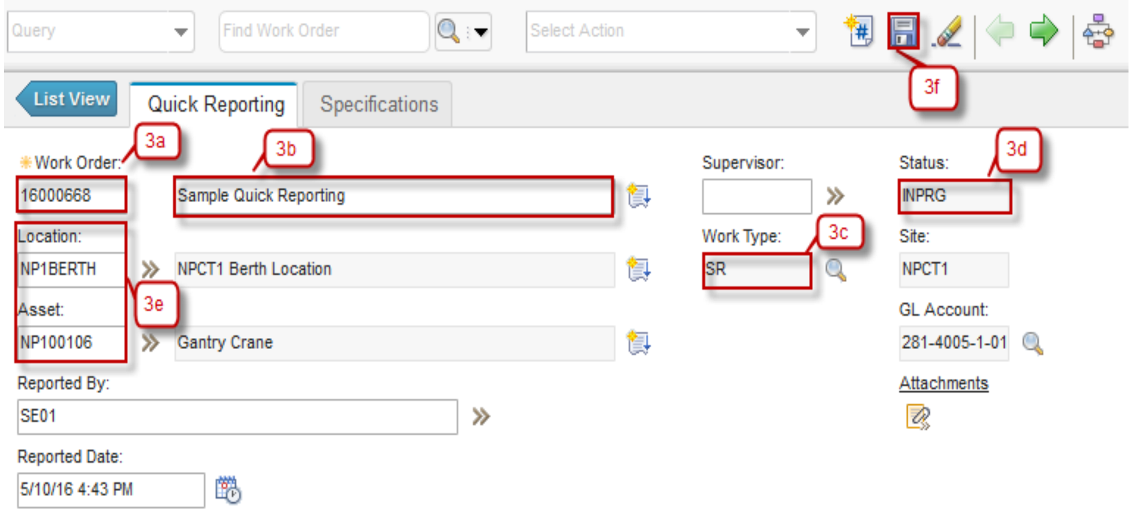
1. SR can be initiated without any prior work order, because it is an emergency. Technician assignment is performed directly. Any material issue from storeroom can be done without any prior approval.
2. After the work is done, *Service Engineer* creates a SR work order as a report of the work in **Quick Reporting** application. *Service Engineer* also include any resources used for this SR.

A closing approval assignment will be sent to *CHE Engineer*, whether the work order can be closed or not.

6.2.1 SR Work Reporting & Submission

After a SR work is finished, Service Engineer should create a work reporting in **Quick Reporting** application. Below are step by step for SR work reporting & submission:

<table>
<tr><th>Step</th><th>Action</th></tr>
<tr><td colspan="2">1. Login as Service Engineer. User name : se01, password : password.</td></tr>
<tr><td>2. Go to Quick Reporting Application by click (GoTo) – Work Orders – Quick Reporting</td><td>
</td></tr>
<tr><td colspan="2">3. Click New Quick Reporting button
3a. Work Order number is autonumber.
3b. Input Work Order Description .
3c. Default Work Type of Quick Reporting is ‘SR’ (Service Recovery).
3d. Default Status of Quick Reporting is ‘INPRG’
3e. Click on Asset or Location field and select one value.
3f. Click Save.</td></tr>
</table>



4. Add a task by following this steps:

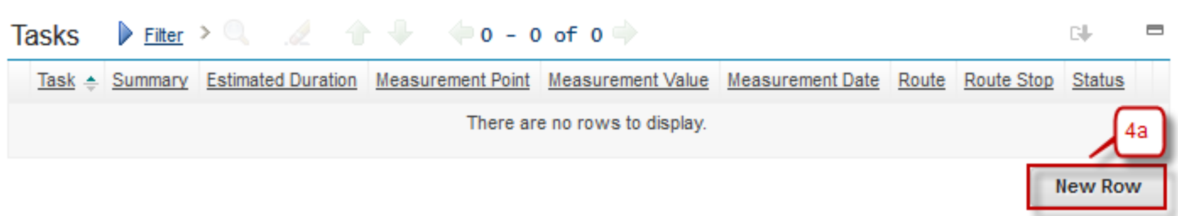
4a. Click **New Row**

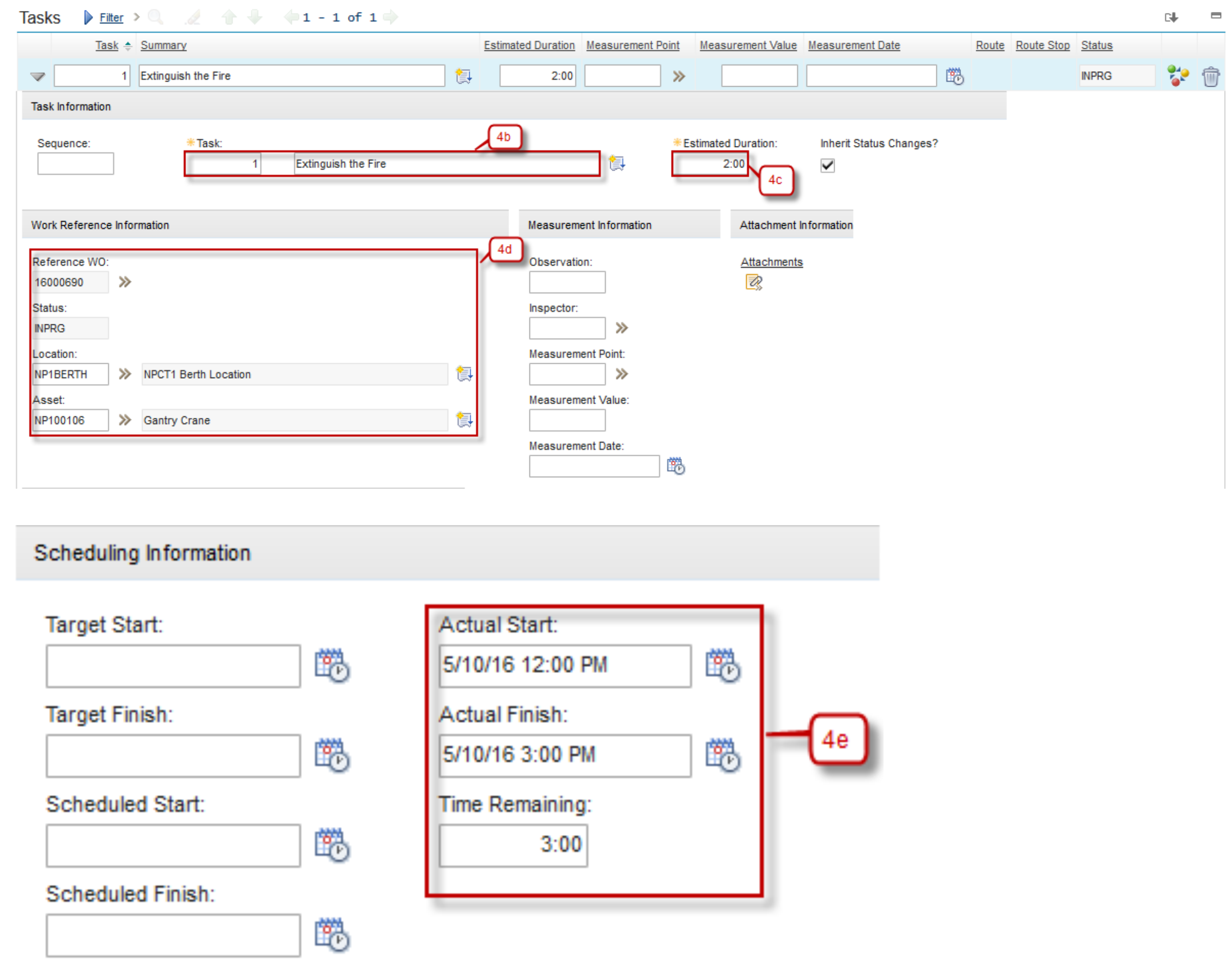
4b. Input Task Description.

4c. Input Estimated Duration.

4d. When inputting task for Work Reference Information Section is automatically filled.

4e. Input the Actual Start and Actual Finish. Value of Time Remaining is Duration minus the number of hours worked so far.





5. Select **Labor** Tab under the page.

5a. Click **Select Labor** button.

5b. Select associated Labor and click **OK**.

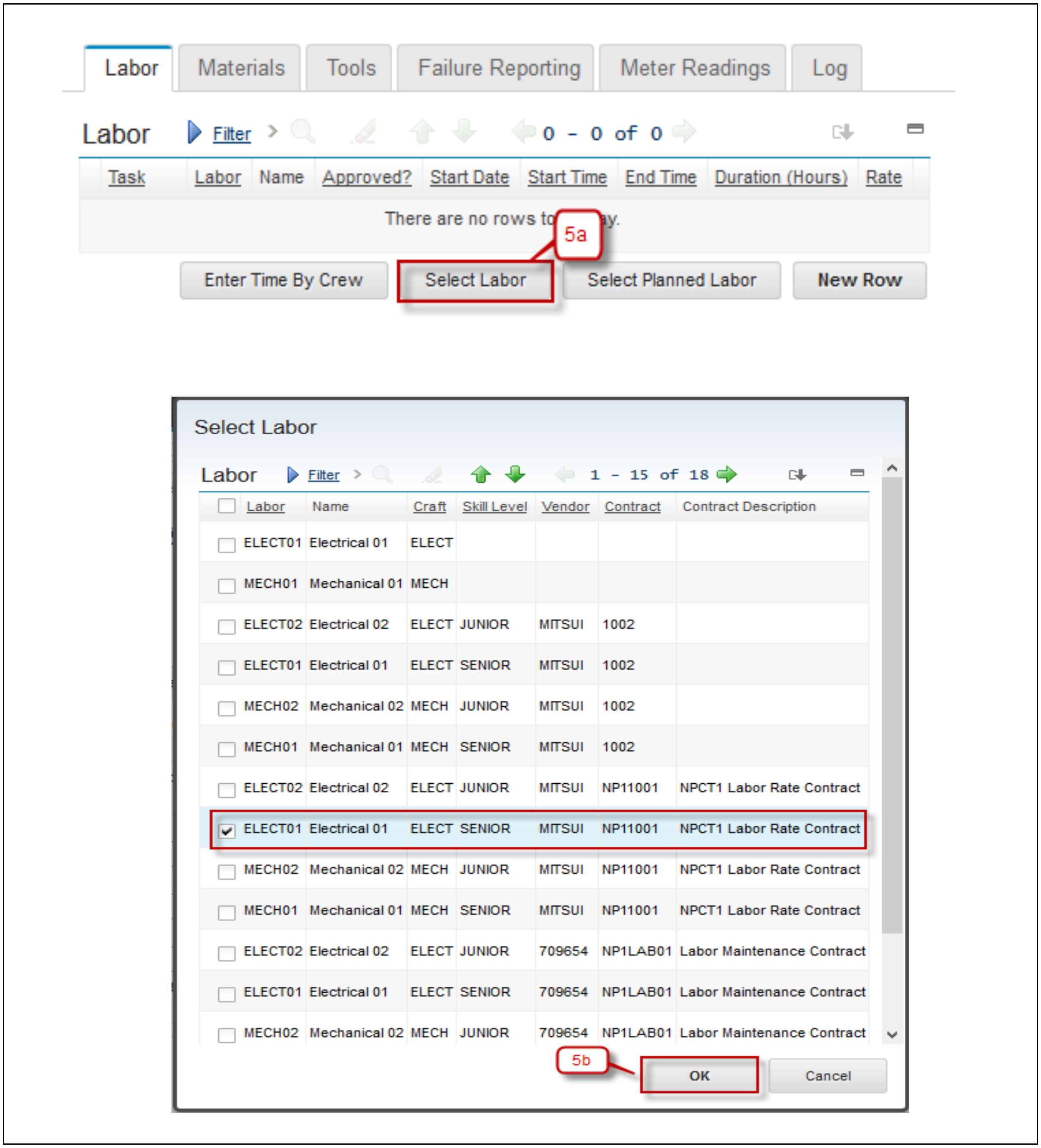
5c. After Select Labor, field Labor, Craft, Skill Level, Start Date, Rate, Type is automatically fill. Input End Date when the task finished.

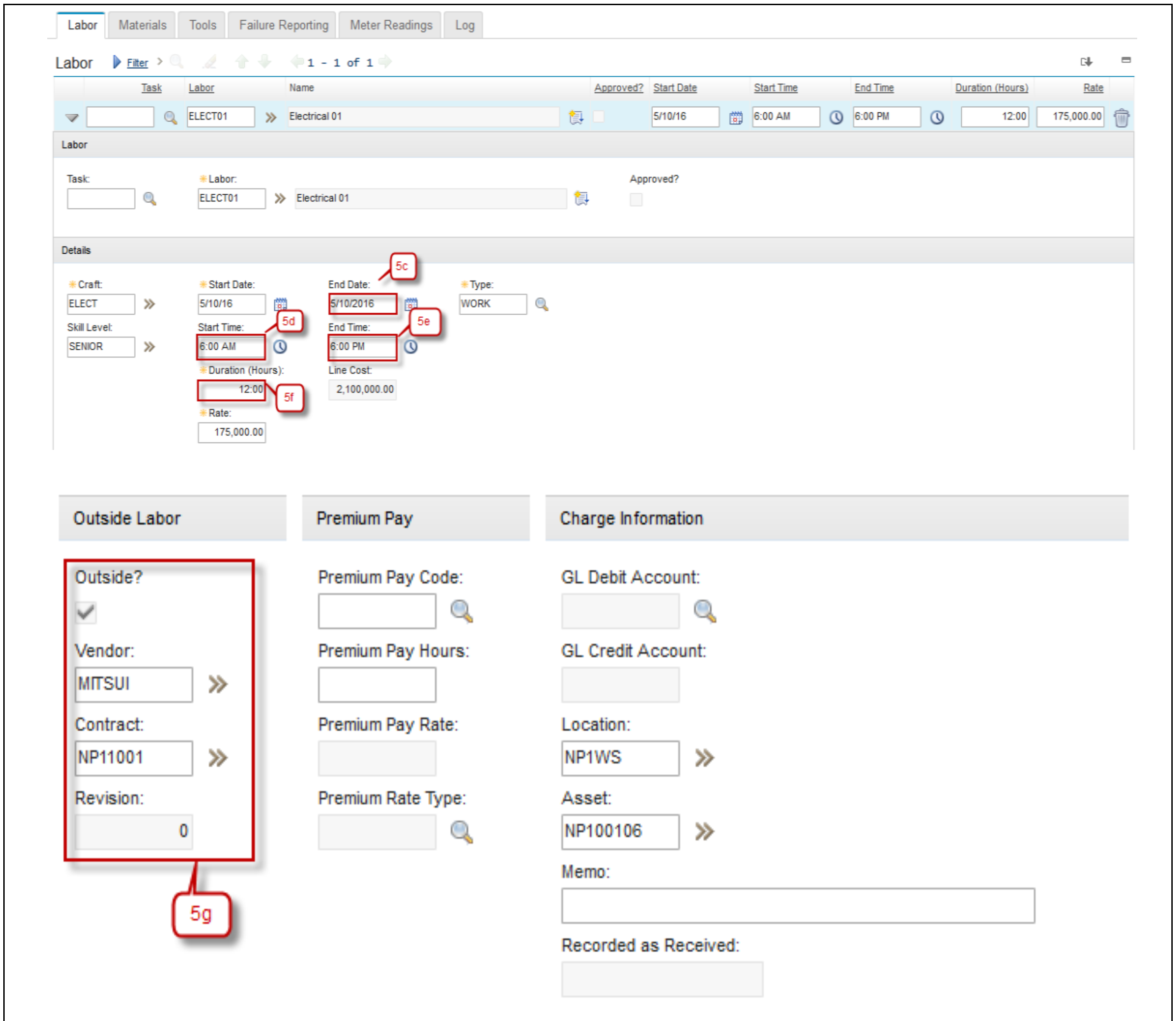
5d. Input Start Time.

5e. Input End Time.

5f. Value of Duration (Hours) is interval between Start Date and End Date.

5g. If Labor using from Outside, the Outside Labor Information will automatically fill.


Labor
Materials
Tools
Failure Reporting
Meter Readings
Log
Labor
Filter
0 - 0 of 0
Task
Labor
Name
Approved?
Start Date
Start Time
End Time
Duration (Hours)
Rate
There are no rows to
ay.
5a
Enter Time By Crew
Select Labor
Select Planned Labor
New Row
Select Labor
Labor
Filter
1 - 15 of 18
Labor
Name
Craft
Skill Level
Vendor
Contract
Contract Description
ELECT01 Electrical 01 ELECT
MECH01 Mechanical 01 MECH
ELECT02 Electrical 02 ELECT JUNIOR MITSUI 1002
ELECT01 Electrical 01 ELECT SENIOR MITSUI 1002
MECH02 Mechanical 02 MECH JUNIOR MITSUI 1002
MECH01 Mechanical 01 MECH SENIOR MITSUI 1002
ELECT02 Electrical 02 ELECT JUNIOR MITSUI NP11001 NPCT1 Labor Rate Contract
ELECT01 Electrical 01 ELECT SENIOR MITSUI NP11001 NPCT1 Labor Rate Contract
MECH02 Mechanical 02 MECH JUNIOR MITSUI NP11001 NPCT1 Labor Rate Contract
MECH01 Mechanical 01 MECH SENIOR MITSUI NP11001 NPCT1 Labor Rate Contract
ELECT02 Electrical 02 ELECT JUNIOR 709654 NP1LAB01 Labor Maintenance Contract
ELECT01 Electrical 01 ELECT SENIOR 709654 NP1LAB01 Labor Maintenance Contract
MECH02 Mechanical 02 MECH JUNIOR 709654 NP1LAB01 Labor Maintenance Contract
5b
OK
Cancel



6. Select **Tools** tab under the page

6a. Click **Select Tools.**

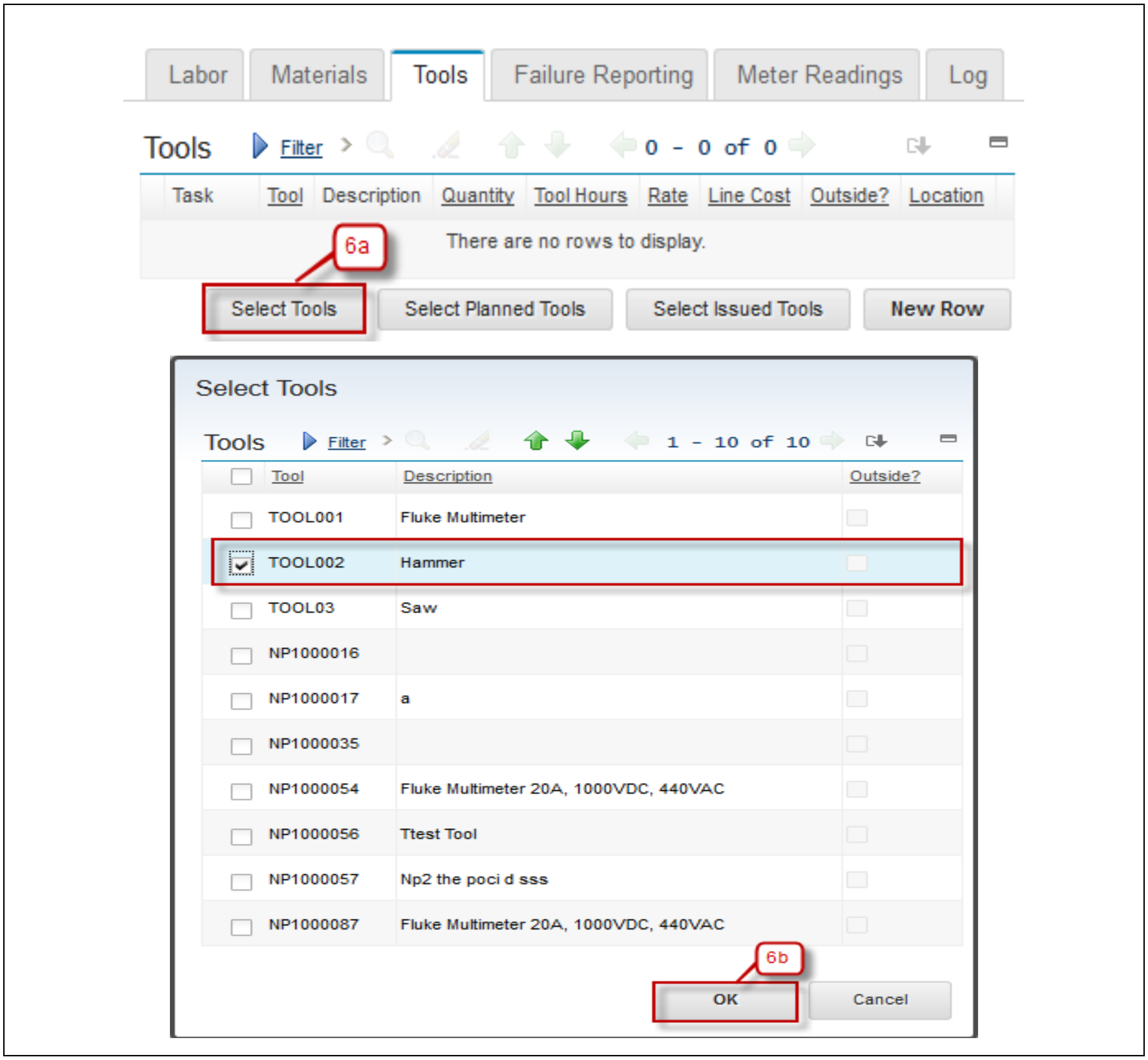
6b. Select tools used for work order and then click **OK.**

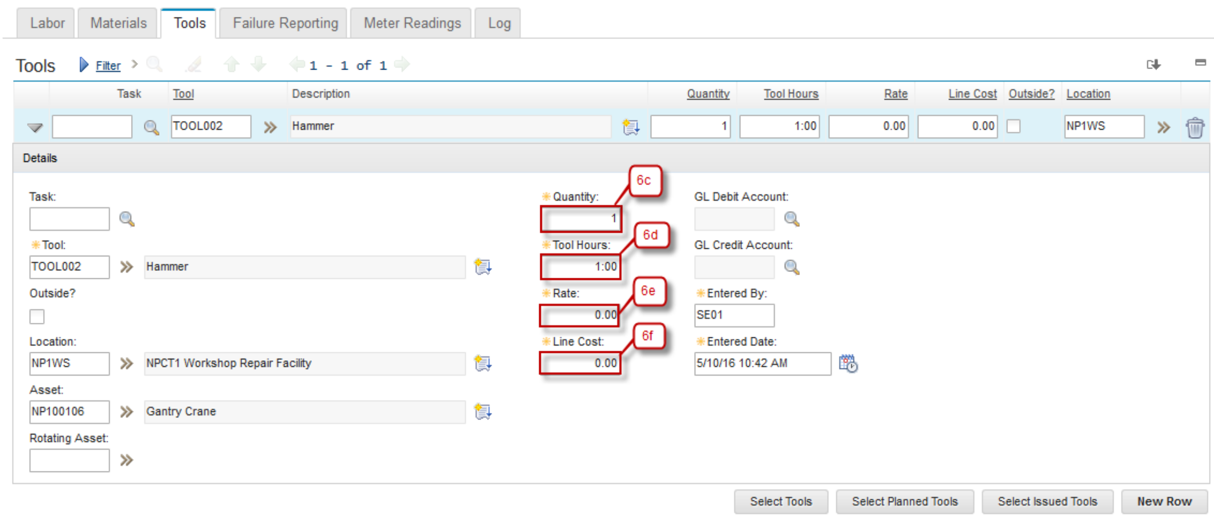
6c. Input Quantity, default value of Quantity is 1.00

6d. Input Tool Hours, default value of Tool Hours is 1.00

6e. Input Rate, default value of Rate is 0.00

6f. Value of Line Cost is multiplication result from Quantity and Rate. Click **Save**

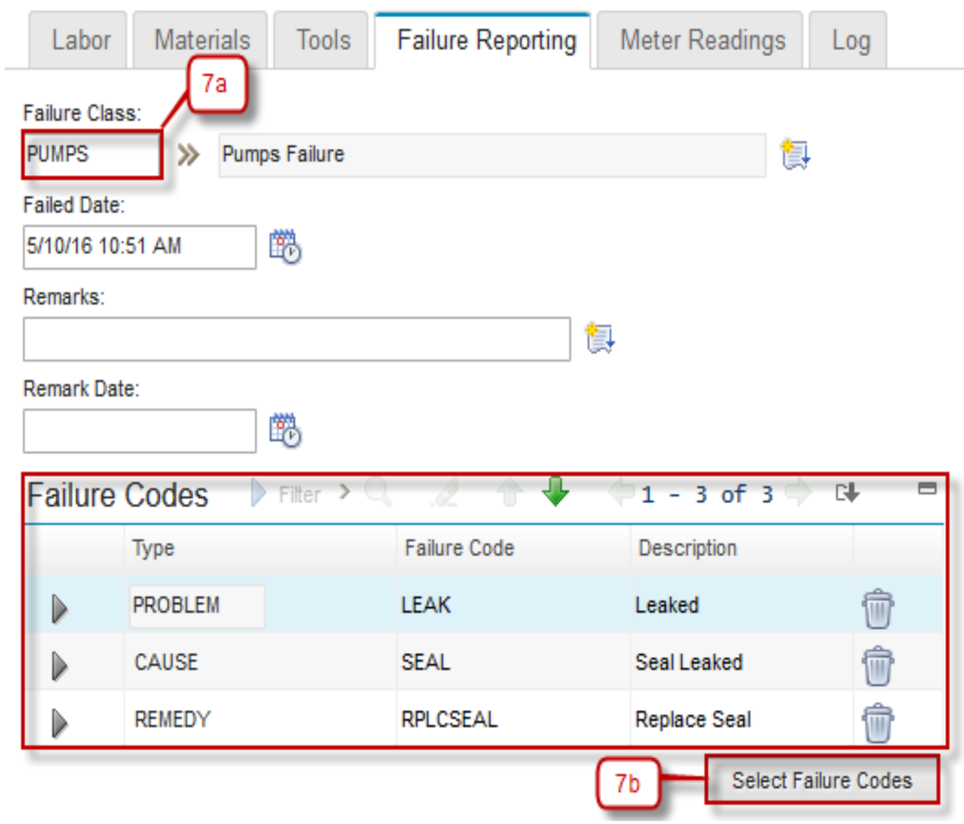

Labor
Materials
Tools
Failure Reporting
Meter Readings
Log
Tools
Filter
0 - 0 of 0
Task
Tool
Description
Quantity
Tool Hours
Rate
Line Cost
Outside?
Location
There are no rows to display.
6a
Select Tools
Select Planned Tools
Select Issued Tools
New Row
Select Tools
Tools
Filter
1 - 10 of 10
Tool
Description
Outside?
TOOL001
Fluke Multimeter
TOOL002
Hammer
TOOL03
Saw
NP1000016
NP1000017
a
NP1000035
NP1000054
Fluke Multimeter 20A, 1000VDC, 440VAC
NP1000056
Ttest Tool
NP1000057
Np2 the poci d sss
NP1000087
Fluke Multimeter 20A, 1000VDC, 440VAC
6b
OK
Cancel



7. Select Failure Reporting tab under the page.

7a. Select Failure Class.

7b. Click Select Failure Codes, dialog box will be appeared and then select failure codes.



8. Complete Quick Reporting

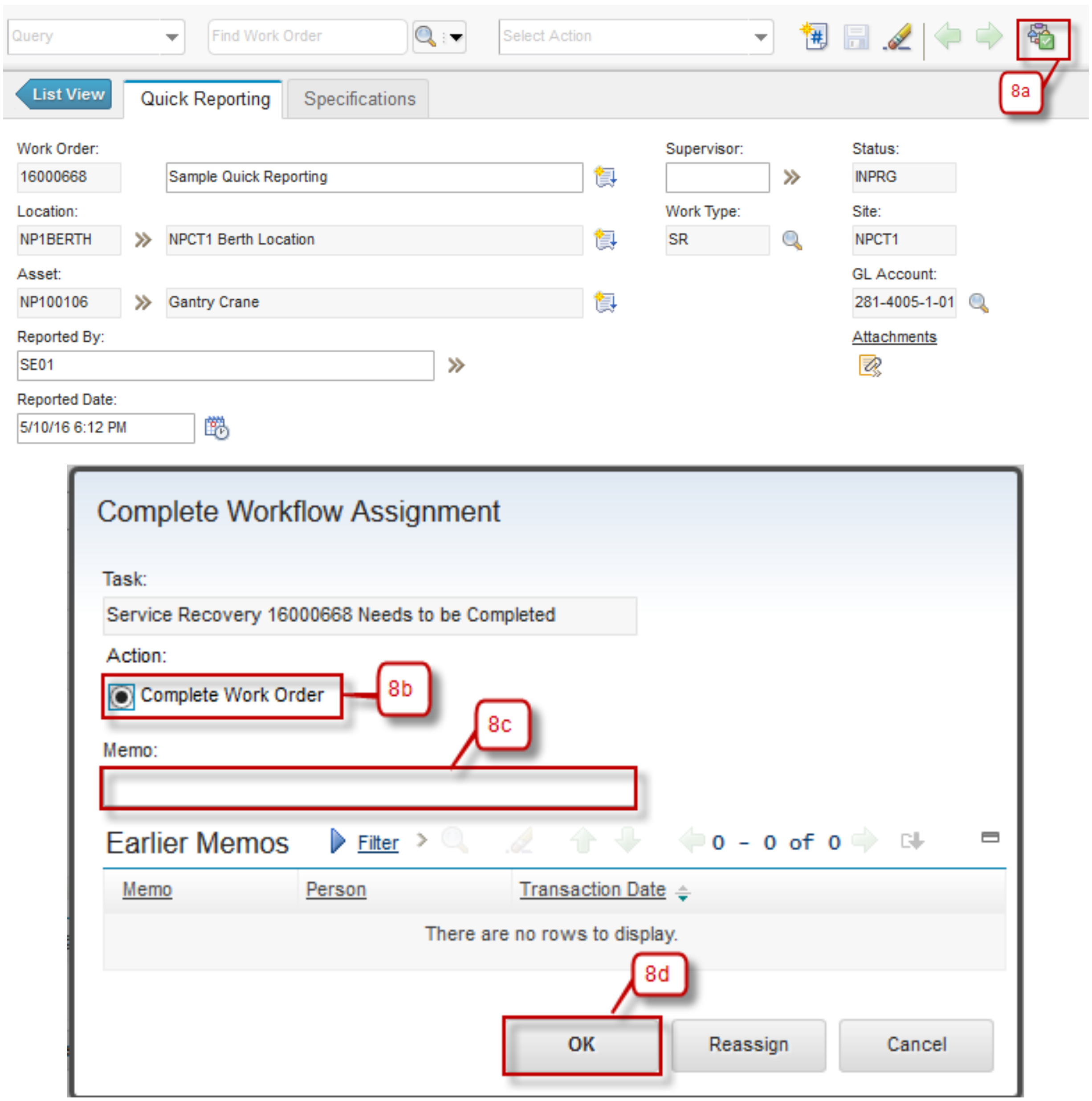
8a. Click  (Route Workflow) button.

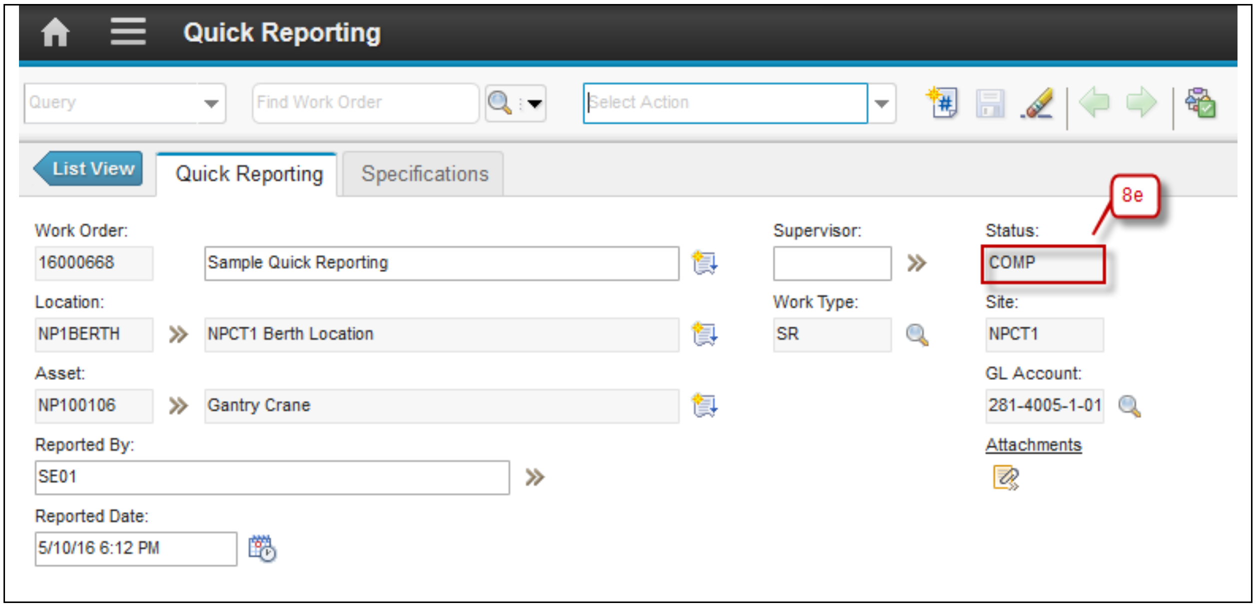
8b. Dialog box **Complete Workflow Assignment** will appear. Choose action **Complete Work Order**.

8c. Input Memo (if needed).

8d. Click **OK.**

8e. Status will be changed to '**COMP**' (**Complete**).



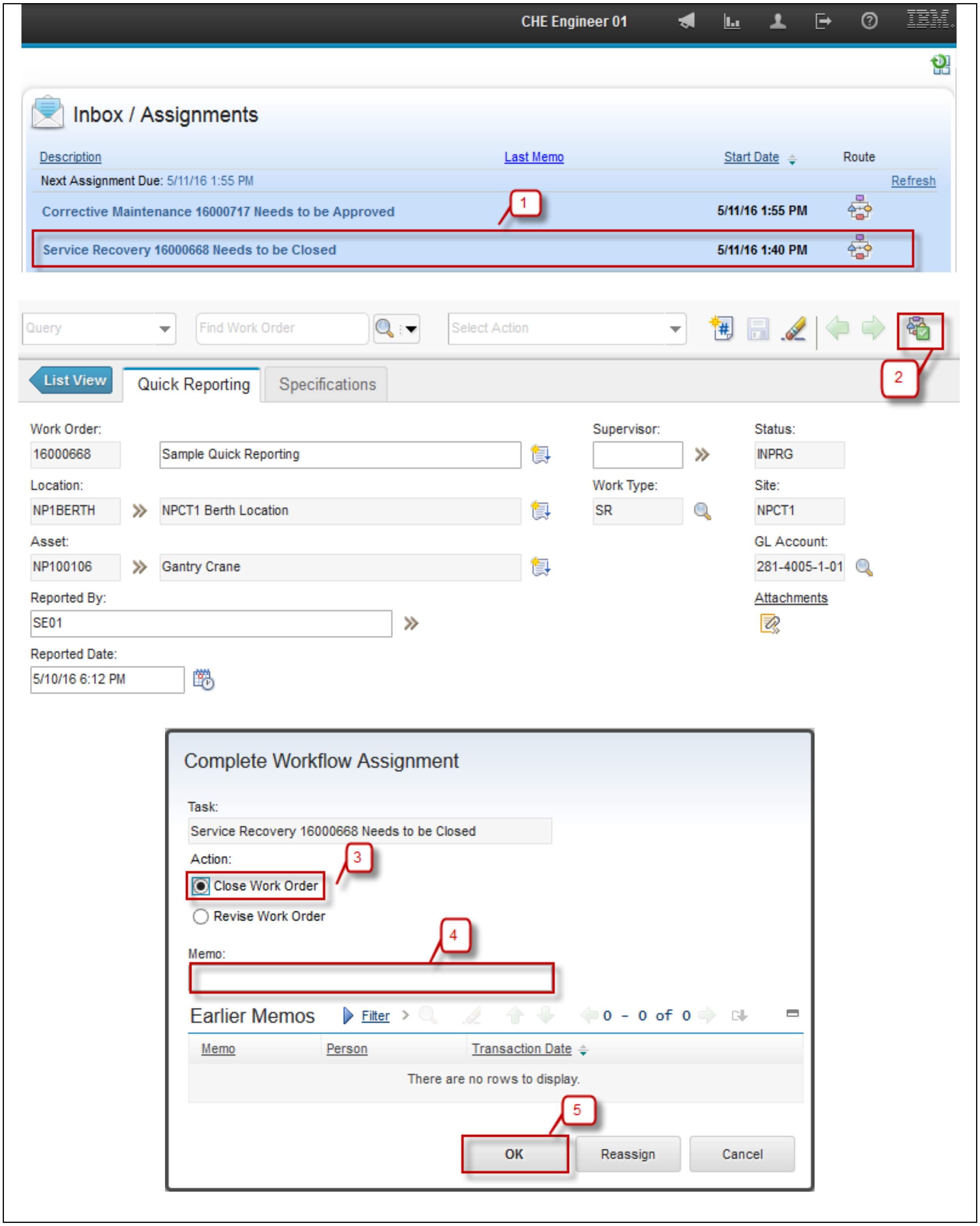


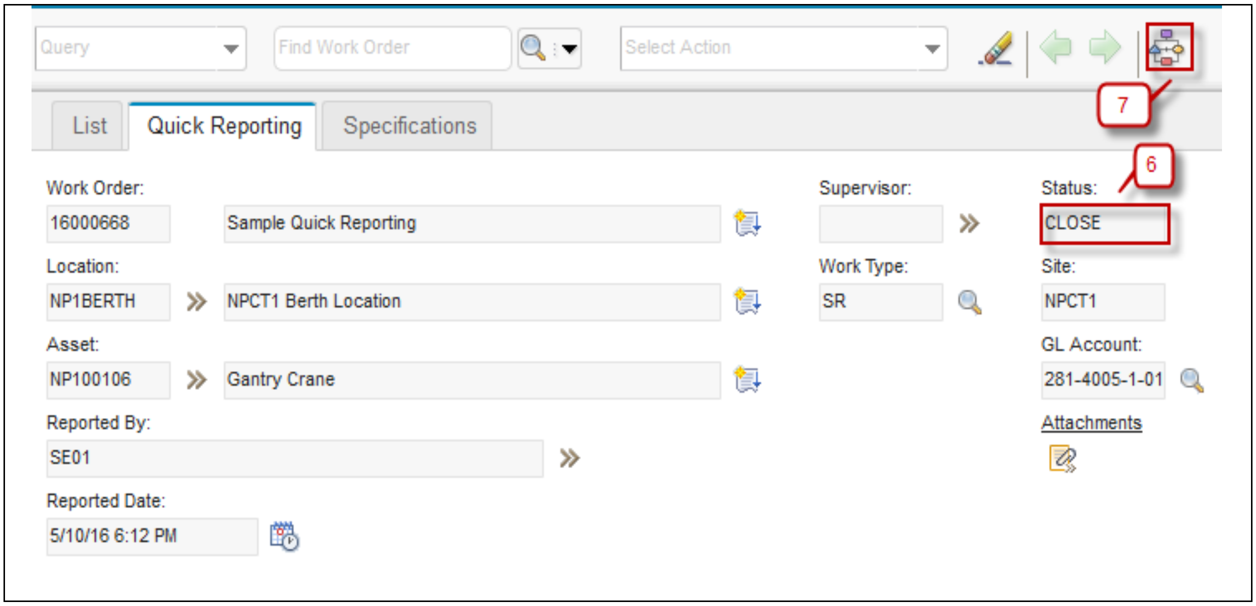
6.2.2 SR Work Closing Approval by CHE Engineer

Below are steps to approve a SR work closing by *CHE Engineer*. For revision request, and revision will not be discussed. They can refer to revision request, and revision of **SM Work Cycle** in step point **1.2.5 SM Request Revision Work Order** and **1.2.6 SM Revise and Re-submit Work Order** training material.

As a note, request revision for SR work closing will direct the assignment back to *Service Engineer*. After revision is done, assignment will be directed back to *CHE Engineer* for closing approval.

Step	Action
1.	Login as CHE Engineer. Username : che01, password: password. On Start Center Operation An assignment to close SR Work Order will appears in **Inbox/Assignment** portlet, click Description of the assignment to go to related Work Order
2.	Click (**Route Workflow**) button and dialog box **Complete Workflow Assignment** will appears.
3.	Choose **Close Work Order.**
4.	Input Memo (if needed).
5.	Then click **OK.**
6.	Status changed to '**CLOSE**'.
7.	The **Route Workflow** button appearance returned back into , indicates that the workflow is finished.


CHE Engineer 01
Inbox / Assignments
Description
Last Memo
Start Date
Route
Next Assignment Due: 5/11/16 1:55 PM
Refresh
Corrective Maintenance 16000717 Needs to be Approved
5/11/16 1:55 PM
1
Service Recovery 16000668 Needs to be Closed
5/11/16 1:40 PM
Query
Find Work Order
Select Action
2
List View
Quick Reporting
Specifications
Work Order:
16000668
Sample Quick Reporting
Supervisor:
Status:
INPRG
Location:
NP1BERTH
NPCT1 Berth Location
Work Type:
SR
Site:
NPCT1
Asset:
NP100106
Gantry Crane
GL Account:
281-4005-1-01
Reported By:
SE01
Attachments
Reported Date:
5/10/16 6:12 PM
Complete Workflow Assignment
Task:
Service Recovery 16000668 Needs to be Closed
Action:
3
Close Work Order
Revise Work Order
4
Memo:
Earlier Memos
Filter
0 - 0 of 0
Memo
Person
Transaction Date
There are no rows to display.
5
OK
Reassign
Cancel



6.3 Exercises